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Energy, exergy, and economic assessment of thermal regulation of PV panel using hybrid heat pipe-phase change material cooling system

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ABSTRACT

There has been a theoretical investigation for energy, exergy, and economic evaluation of thermal regulation of PV panels via an innovative hybrid cooling system of flat heat pipes (HPs) coupled with phase change material (PCM). PV panel dissipates its heat through the HP evaporator while the HP condenser is cooled by PCM. The cooling system performance is investigated for different PCM quantities and types (RT42, RT35HC, and SP31) for summer and (RT21, RT18HC, and SP15-gel) for winter conditions in Egypt. A thorough study and analysis have been conducted for the impact of the cooling system on PV panel's temperature, electric efficiency, output power, PCM energy storage, and system efficiency. The results show that compared with conventional PV, the best PCM type for PV performance in summer is SP31 of 3 cm thickness, which achieves a maximum PV temperature reduction of 20.6 °C with a maximum improvement in PV electric efficiency of 11.5%. SP15-gel of thickness 2.5 cm is the best PCM for winter conditions, achieving 14.1 °C temperature reduction with 7.2% enhancement in PV electric efficiency. The exergy efficiency of the PV panel is higher than in PCM, where it rises as the phase change temperature of the PCM increases. PV maximum average daily exergy efficiency is 13.4% for SP15-gel, compared with 12.71% for conventional PV in winter, while SP31 achieved 12.25%, 1% higher than reference PV in summer. PCM's maximum average daily exergy efficiency is 0.69% and 0.7% for RT42 and RT21 in summer and winter, respectively. The minimum payback period for conventional PV is about 18.01 years, while it is approximately 13.39 years for PV with HP and PCM. The studied cooling system at different PCMs proves its capability to enhance the PV performance with higher energy and exergy efficiencies and lower cost and store the excess heat inside the PCM.

1. Introduction

Recently, the demand for fossil fuels has increased dramatically, particularly for electrical energy, resulting in a significant depletion of these resources (Su et al., 2018). Renewable energy technologies have been at the forefront of technological innovation in the energy sector to address environmental sustainability, rising fuel prices, and the depletion of non-renewable resources (Sharma et al., 2017). A significant amount of potential has been identified in solar energy, including photovoltaics (PV) and photovoltaic thermal collectors (PVT) (Jamil et al., 2021). PV panels directly convert solar radiation into electricity; however, almost 60% of the sun's rays are converted into waste heat, raising the cell's operating temperature (Duan, 2021). The high operating temperature of the solar cells declines the output power by approximately 0.65–0.8% per degree Celsius of the temperature increase and causes irreversible damage to the cell materials (Abdulmunem et al., 2021). It is necessary to remove the excess heat from the PV panels using an appropriate method to alleviate these undesirable effects. These techniques must achieve particular objectives, including being easy to install, low operating cost, low thermal resistance, and maintaining uniform cell temperature.

Cooling methods of PV can be broadly classified into active and passive approaches. In the case of active cooling, pumps or fans are used to drive the coolant flow. The use of circulated air or water is a simple and natural method of accomplishing this goal (Bhakre et al., 2021).

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Nomenclature

A	Area (m^2)
AS	Annual net cost saving
C	Specific heat capacity ($J{\cdot}kg^{-1}\ K^{-1}$)
D	Width (m)
E_{annual}	Annual electric power output
$E_{th,latent}$	Absorbed latent heat
G	Solar radiation ($W{\cdot}m^{-2}$)
g	Gravity acceleration ($m.s^{-2}$)
h	film coefficient ($W{\cdot}m^{-2}\ K^{-1}$)
h_a	convective heat transfer coefficient ($W{\cdot}m^{-2}\ K^{-1}$)
$h_{rad(s-g)}$	radiative heat transfer coefficient ($W{\cdot}m^{-2}\ K^{-1}$)
h_{fg}	Vaporization heat ($J{\cdot}kg^{-1}$)
I	Interest rate (%)
k	Thermal conductivity ($W{\cdot}m^{-2}\ K^{-1}$)
L	Length (m)
LH	Latent heat ($J{\cdot}kg^{-1}$)
M	Mass (kg)
n	Number of years
Nu	Nusselt number
P	Power (W)
P_e	Electricity price ($\$\cdot k^{-1}\ W^{-1}\ h^{-1}$)
Pr	Prendle number
R	Thermal resistance ($K{\cdot}W^{-1}$)
R_a	Rayleigh number
r	Interest rate (%)
T	Temperature (°C)
T_o	environment temperature (K)
T_{sun}	Solar radiation temperature (K)
t	Time (s)
u	Velocity ($m.s^{-1}$)
w	groove width (m)
in	Input
ins	Insulation
l	Liquid
m_1	Melting start point
m_2	Melting endpoint
rad	Radiation
ref	Reference
s	Sky, Solid
th	Thermal
t_o	Initial time
v	Vapor
ve	liquid evaporation
wa	Heat pipe wall

Abbreviation

AMC	Annual maintenance cost
AS	annual net cost savings
ASV	annual salvage value
E	Energy (J)
Ex	Exergy
$E_{th,sen,l}$	Sensible heat storage for liquid
$E_{th,sen,s}$	Sensible heat storage for solid
w_f	groove fin thickness (m)
x	Thickness (m)
ΔT_{vc}	Temperature difference between fluid -and heat pipe wall

Greek symbols

β	Temperature coefficient (k^{-1})
α	Absorptivity (%), thermal diffusivity ($m^2{\cdot}s^{-1}$)
σ	Stefan–Boltzmann constant ($W{\cdot}m^{-2}\ K^{-4}$)
δ	Groove depth (m)
ε	Emissivity
θ	Inclination angle(°)
φ	Porosity (%)
η	efficiency (%)
λ	Liquid fraction (%)
ρ	Density ($kg{\cdot}m^{-3}$)
τ	Transmittance
ζ	Exergy efficiency (%)
μ	Dynamic viscosity ($N{\cdot}s{\cdot}m^{-2}$)
ν	Kinematic viscosity ($m^2{\cdot}s^{-1}$)

Subscripts

a	Ambient, Air
avg	Average
c	Characteristic
con	Condenser section
cond	Vapor condensation
e	Electric
ev	Evaporator section
g	Glass
CC	capital cost
CRF	capital recovery factor
EVA	Ethyl vinyl acetate
FAC	first annual cost
HP	Flat heat pipe
HS	Heat sink
HP-PCM	Integrated flat heat pipe-phase change material
PB	Payback period
PCM	Phase change material
PV	Photovoltaic
PV/HP-PCM	Integrated photovoltaic/flat heat pipe-phase change material
PV/PCM	Integrated photovoltaic-phase change material
PVT	Photovoltaic thermal
S	Salvage value
SFF	sinking fund factor
TAC	Total annual cost
TPT	Tedlar/PET/Tedlar

However, active cooling is solely used to lower the operating temperature and does not provide a significant net gain in efficiency because of the increased power consumption (Wu and Xiong, 2014). To avoid the need for additional energy input (electrical or mechanical power) and to eliminate moving media (Sharma et al., 2017), a uniform cooling (passive cooling) using phase change materials (PCM), extended surfaces (fines), heat sinks and spreaders, and heat pipes (HP) has been reported by many researchers.

PCMs are latent heat storage materials in their ability to store heat. PCMs are used in a wide range of thermal engineering applications, such as air-conditioning cycles (Said and Hassan, 2018), the absorption cycle (Soliman et al., 2021a), generation mechanical work from PCM expansion for driving a solar cell tracking system (Mendecka et al., 2022), thermal energy storage in building to balance the demand for cooling and heating, waste heat, and reducing mechanical stress, deformation, and cracks on roadways (Borri et al., 2021). Because of their high thermal capacity, PCMs have been the subject of extensive research in the solar energy field (Pandey et al., 2018). The excess heat generated by PV cells can be absorbed by the PCM, which is attached to the back of the PV module. Integrated photovoltaic with phase change material (PV/PCM) is a hybrid technology in which the PV panel and the PCM storage are coupled into a single component to achieve higher electrical

conversion efficiency and lower PV surface temperature. During sunny hours, when the operating temperature of the PV panel is higher than the melting point of the PCM, the PCM storage acts as a heat sink for the PV panel's dissipated heat, maintaining the PV panel's temperature near its melting point without additional energy input (Waqas et al., 2018). In this way, the PV panel is passively cooled, and the conversion efficiency is improved without using any parasitic energy from the PV panel. For different weather conditions (Savvakis et al., 2020), used two PCM materials (RT27 and RT31) to control the PV temperature. They illustrated that using 0.26 kg of PCM RT27 and PCM RT31 decreases the maximum module temperature by 6.4 °C and 7.5 °C compared with PV only, respectively. Also, they reported that the increase in the electric efficiency of PV/PCM RT31 was 4.19%, while PV/PCM RT27 saw a rise of 2.86%. Finally, they reported that decreasing PV temperature by one °C required 340 g/m^2 of PCM; however, the suitable amount of PCM differs according to its properties. Using different layers of PCM with a suitable arrangement and configuration increases the period of the thermal regulation of the PV compared with using only one PCM type (Mahdi et al., 2021).

Building-integrated photovoltaic thermal (BIPV/T) systems have had a significant interest recently. In this system heat-absorption material is used to cool down the PV cell and store heat for other purposes. In BIPV/T PCM, PCM is placed behind PV to reduce the cell temperature and store absorbed heat for heating applications (Sohani et al., 2022). In their study, the mathematical model was solved using MATLAB software, using dynamic multi-objective optimization (MOO) to determine the optimum PCM thickness. They reported that 7.72 cm was the best thickness of PCM, achieving 22.24% and 9.93% enhancement in annual thermal energy storage and electricity production, respectively. Another study was performed by (Kant et al., 2020), where they used nanoparticles to enhance the thermal conductivity of PCM integrated with BIPV (BIPV/NPCM). In this study, the mathematical model and simulation were performed using the commercial package of COMSOL. The proposed system kept the operation temperature of the cell under 40 °C for 1 h under solar radiation of 500 W/m^2.

A few studies reported the effect of PCM type and amount on PV panel thermal and electric performance (Ho et al., 2016). developed a simulation analysis to study the effect of PCM amount and melting point for different weather conditions on the performance of the PV system. They studied a range of melting points (26 °C–30 °C) in summer and (16 °C–18 °C) in winter with PCM thicknesses (3 and 5 cm). Maximum performance was obtained using a PCM arrangement with a melting point of 30 °C/26 °C with a layer thickness of 5 cm, leading to a reduction of 0.9 °C in photocell temperature in summer. In winter, using 5 cm thickness with a melting point arrangement (18 °C/16 °C) achieved a reduction of 2.9 °C. Excess heat can be removed and used for other purposes to reduce the impact of overheating on the panel using the PVT approach. Also, utilizing extracted heat improves the overall efficiency and cost-effectiveness of PV/PCM. To maximize the energy output and enhance the performance of the PV/T solar collector (Su et al., 2017), numerically studied the effect of PCM amount and melting point on the electric and thermal performance of the PV/T collector. Their findings revealed that a PV/T collector provides the optimum electrical performance with a PCM melting point of 30 °C, and the optimum thermal energy gain was accomplished at a PCM melting point of 40 °C with 34 mm thickness. Enhanced Conduction Model (ECM) was developed by (Elsheniti et al., 2020) to predicate the performance of PV regulated thermally by PCM for different inclined angles. Different PCM thicknesses (10 mm–40 mm) with melting ranges (25 °C–26.5 °C) were used for winter conditions. They reported that PV/PCM performed efficiently with a PCM thickness of 30 mm. Heat stored in PCM can also be utilized in electricity generation and providing. For example, electricity can be generated around the clock in summer and winter (Naderi et al., 2021). They reported that the temperature of the solar cells was reduced by about 20 °C, and the system's electricity output and solar cell efficiency were both increased by 100% and 1.38%, respectively.

The low thermal conductivity of PCMs is the most significant issue associated with them, as it causes the heat transfer rate to be retarded and thus limits their practical applications. However, among different enhancement techniques, heat pipe (HP) has superiority in transporting heat at high rates through a confined area to considerable distances with a significantly lesser temperature gradient (Rashidi et al., 2020). HP provides several advantages and operational and application flexibility, such as easy control, powerful thermal performance, simple construction, compact structure, changeability to a variety of layout conditions, and no external pumping power is required (Putra et al., 2020). Many studies have investigated the possibility of using heat pipes to extract and transfer excess heat from PV cells isothermally over both short and long distances (Soliman and Hassan, 2020). Recently, heat pipes have been found in a wide range of photovoltaic thermal management and PVT systems to achieve high electric and thermal gain by extracting heat from PV panels with heat pipes (Hassan and Harmand, 2013a; Huang et al., 2021).

A potential solution to the poor thermal conductivity of PCM is to use HPs as a heat transfer medium between the heat source and the PCM storage device. Combining a high thermal conductivity HP with PCM's high heat capacity is the primary goal of an integrated approach (Behi et al., 2021). Besides regulating the temperature, PCM-assisted heat pipe modules also help prevent HP from overheating and improve PCM thermal performance. Numerous studies in various applications have advocated for integrated heat pipes with phase change materials (HP-PCM) as a cooling option (Behi et al., 2021). used flat heat pipe with the assistance of PCM to extract heat from electric vehicle batteries. They reported that the maximum temperature of the battery using HPS-assistant PCM was 33.2 °C, compared with 46.3 °C using HPs only. The hybrid HP-PCM is also used for cooling electronic devices where heat pipes were used to extract heat from electronic devices and PCM as a storage unit for extracted heat (Krishna et al., 2017). The feasibility of using a hybrid HPs-PCM for thermal regulation of a PV/T system was proposed by (Sweidan et al., 2016). In this work, a water tank with cylindrical PCM tubes was used to extract heat from the heat pipe condenser, where water was used to cool the heat pipe, and part of the heat, which was absorbed by water, was stored in PCM. They reported that for 4 kW power output, the optimal volume of the storage tank is 37 l with 22.42 kg of PCM at a melting point of 33 °C. A design for PV/PCM/heat pipe was proposed by (González-Peña et al., 2020). In this design, a PCM layer was added to the PV module's back surface to control the module's operating temperature by absorbing excess heat. Moreover, the outer back edges of the PV module were black surface to increase absorbed heat transported to PCM using heat pipes.

The use of exergy-based methodologies to evaluate and optimize various solar energy systems establishes a clear link between the system and the environment, providing an understanding of the system's environmental impact. Few scholars investigated the exergy analysis of the PV system. 4E assessment of PV integrated with PCM for thermal regulation studied by (Yousef et al., 2022). They reported PV's thermal and electric performance without a detailed analysis of PCM. The results revealed that when compared to the conventional PV setup, the electrical efficiency of PV-PCM/AF and PV-PCM systems was increased by 14% and 9%, respectively, in summer. These values in winter were 4.8% and 3.7%, respectively. Also, they performed electric exergy analysis for PV only. The results showed that the average exergy efficiency for the modified system for summer and winter conditions was 1.58% and 0.6% higher than the conventional PV, respectively. Exergy analysis is a straightforward method to determine the amount of energy available for productive purposes (Eisapour et al., 2021). carried out theoretical energy and exergy study for the PVT system. In that study, they used different coolant tube configurations and coolant fluid. Due to the improved distribution of the cooling tube, lower PV temperatures can be achieved by utilizing ascending wavy tubes with enhancement in exergy, electric, and thermal efficiencies of 12.34%, 11.32%, and 65.21%, respectively, compared with 11.89%, 10.94%, and 61.04% for

the straight tube. Wahab et al. (2020) performed an exergy analysis for hybrid PV integrated with PCM for passive cooling and water with nanofluid for active cooling. Their results revealed that compared to electrical exergy, the impact of thermal exergy on overall exergy efficiency was minimal, with maximum overall exergy efficiency being 14.62% achieved at a water flow rate of 40 l/m with 0.1% volume concentration of nanoparticles. It is clear from previous studies that thermal exergy efficiency is minimal compared with overall exergy efficiency.

According to the literature, HPs can be used in conjunction with a heat source and a PCM storage unit to compensate for the low thermal conductivity of PCM for cooling various thermal systems, such as electric batteries and electronic devices. Some studies concentrated on thermal regulation of PV using heat pipe; however, relatively few studies have been conducted on the potential of the hybrid HP-PCM system for solar cell applications. To the best of the authors' survey, studying the performance of coupling the flat heat pipe with phase change materials has not received attention in the literature, especially regarding thermal regulation of PV panels as a passive cooling technique (using HPs to absorb excess heat from PV cell and transport it to be stored in PCM). Moreover, most studies evaluating PV thermal regulation were energy-based; however, few studies with exergy-based or economic-based assessments can be found in the literature. Therefore, the current study investigates the performance of the hybrid HP-PCM cooling technique for thermal regulation of the PV panels based on energy, exergy, and economical approach. The PV represents the heat pipe evaporator's heat source while the PCM, which works as a heat sink, cools down the condenser section. This study is performed at different amounts and types of PCM (RT42, RT35HC, and SP31) and (RT21, RT18HC, and SP15-gel) for summer and winter climate conditions, respectively, in Assiut city, Egypt. A transient mathematical model of all the system components is constructed, programmed, and solved using MATLAB software and validated using theoretical and experimental data from the literature.

One of the advantages of such a cooling system is that it extracts the dissipated heat from the PV system far from it by using a simple passive technique without requiring any power and maintenance. Also, it stores the extracted heat from the photovoltaic system within the PCM unit for use as supplemental heating. Additionally, it increases the PCM storage capacity by controlling its quantity compared with PCM's direct contact with the PV back surface. The effect of the hybrid cooling system on the following is analyzed and reported: the PV panel's thermal and electrical performance, PCM thermal performance, and PV and PCM exergy efficiencies. Also, The best type and amount of PCM are investigated for PV temperature reduction and maximum system efficiency. Finally, the impact of this cooling system on the output power price and cash flow is studied and compared with the natural cooling of the PV.

This paper presents the following points, which presented a gap in the research to the authors' best survey:

- The use of flat heat pipe-phase change material (HP-PCM) for cooling PV under hot and cold climate conditions
- The assessment of energy, exergy, and cost-efficiency for the PV/HP-PCM system
- The impact of PCM type on the system performance
- The impact of PCM quantity on the thermal regulation of the PV panel and PCM thermal performance

The importance of this study lies in the fact that it presents a passive efficient cooling system for PV thermal regulation. Secondly, it provides an assessment of the PV/HP-PCM system from energy, exergy, and economical perspectives.

2. Physical model

There are three main components in the present physical system: PV panel, HP, and PCM container, as shown in Fig. 1. The evaporator section of the heat pipe is attached to the back of the PV cell to absorb excess heat from the PV cell for cooling and thermal regulation. The lower part of the heat pipe, the evaporator section, absorbs heat from the PV cell, enabling its working fluid (acetone) to evaporate. A pressure difference is created whenever heat is applied; thus, causing the vapor to be drawn toward the upper condenser section. When the vapor reaches the condenser section, it condenses and releases the latent heat to the cooling media. The liquid is then drawn back into the pipe from the condenser section by the capillary wick effect (Hassan and Harmand, 2013b). The condenser section of the heat pipe is placed in the middle of the PCM container for better heat dissipation from the heat pipe and more efficient melting of the PCM, as shown in Fig. 1. PCM is contained in an adiabatic box for storing excess heat from the PV cell during the daytime. PCMs work as heat sinks to extract heat from the condenser section of the HP for thermal management of the PV cell; hence, the extraction rate of the PCM for the heat is higher because it depends on the latent heat of PCM melting. Furthermore, the PCM acts as a heat storage unit throughout this operation.

In the presented system, solar radiation, ambient temperature, and wind velocity for a summer and winter day are used in the simulation. The cell directly converts about 20% of solar radiation while absorbing the remaining heat, leading to a high cell operating temperature (Browne et al., 2016). While heat is transferred from the top surface of the PV cell by both convection and radiation, the bottom side is cooled through absorbed heat by the evaporator section of the heat pipe. The current study has evaluated the average PV temperature, liquid fraction, and temperature distribution through PCM using different amounts and types of PCMs. Additionally, for the optimum thickness of PCM, the following elements have been studied: instantaneous electric and thermal efficiencies, instantaneous electric and thermal energy, and average electric and thermal efficiencies. Moreover, exergy and cost analyses are performed on the hybrid PV system. The results of the hybrid cooling system are compared with PV performance under conventional cooling.

3. Mathematical model

A transient mathematical model based on energy balance is used to study the performance and thermal regulation of the proposed module. The governing equations are divided into four parts: the thermal and electric models of PV, the thermal module of the heat pipe, the thermal model of phase change material, and economic analysis.

3.1. Thermal energy balance for PV layers

In sunlight, some energy is lost to the atmosphere through radiation and convection. At the same time, the remaining part of solar radiation is absorbed by the PV panel, resulting in the generation of heat and electrical energy. The heat generated is transferred to the heat pipe evaporator section through conduction from the lower side of the PV panel. As shown in Fig. 1, the PV panel is constructed from five layers; glass cover, upper encapsulated ethyl vinyl acetate (EVA), silicon cell (polycrystalline silicon), lower EVA, and TPT.

3.1.1. Energy balance of glass cover

The energy balance of the glass cover includes the heat transfer by convection to ambient $[h_a A_g(T_g - T_a)]$, heat transfer by radiation to sky $[h_{rad(s-g)} A_g(T_g - T_s)]$, heat transfer by conduction to EVA layer $\left[\frac{T_g - T_{EVA1}}{R_{g-EVA}}\right]$, and the rest of solar energy absorbed by the solar cell after transmission $[GA_g\alpha_g]$. The transient energy balance of the glass cover can be represented by Eq. (1)

$$M_g C_g \frac{\partial T_g}{\partial t} = GA_g\alpha_g - h_a A_g\left(T_g - T_a\right) - h_{rad(s-g)} A_g\left(T_g - T_s\right) - \frac{T_g - T_{EVA1}}{R_{g-EVA}} \quad (1)$$

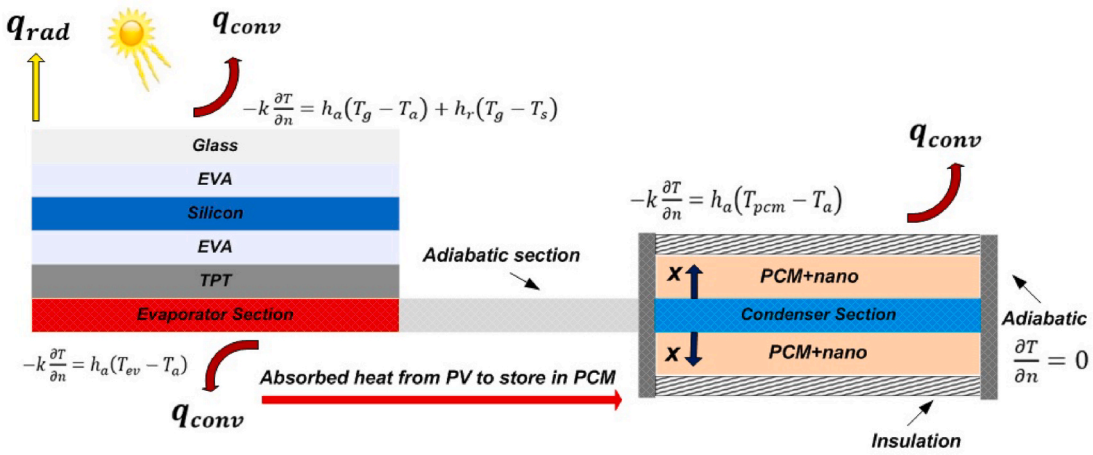


Fig. 1. Schematic diagram of the PV/HP/PCM system evaluated in this study.

the convective (h_a) and radiative ($h_{rad(s-g)}$) heat transfer coefficients for a glass cover exposed to the ambient environment, as shown in Fig. 1, can be calculated using Eqs. (2) and (3), respectively (Notton et al., 2005).

$$h_a = 5.82 + 4.07u_a \tag{2}$$

$$h_{rad(s-g)} = \varepsilon_g \sigma \left(T_g^2 + T_s^2\right)\left(T_g + T_s\right) \tag{3}$$

T_s is the sky temperature, with $T_s = T_a - dt$ (Notton et al., 2005), with T in K and dt is given as follows:

$dt = 20\ k$ for clear sky conditions (summer).

$dt = 0\ k$ for cloud sky conditions (winter).

3.1.2. Energy balance of upper EVA

$$M_{EVA}C_{EVA}\frac{\partial T_{EVA1}}{\partial t} = GA_{EVA}\alpha_{EVA}\tau_g + \frac{T_g - T_{EVA1}}{R_{g-EVA}} - \frac{T_{EVA1} - T_{pv}}{R_{pv-EVA}} \tag{4}$$

3.1.3. Energy balance of silicon cell

The time-dependent temperature variation for silicon layers consists of the amount of solar energy transmitted from the above layers and absorbed by the silicon layer $[GA_{PV}\ \alpha_{pv}\ \tau_g\tau_{EVA}]$, the rate of the electrical energy available from the PV module $[P_{pv}]$, and rate of heat transfer by conduction to and from the silicon layer $\left[\frac{T_{EVA1}-T_{pv}}{R_{pv-EVA}} - \frac{T_{pv}-T_{EVA2}}{R_{pv-EVA}}\right]$. The energy balance equation of the silicon layer is given by Eq. (5).

$$M_{pv}C_{pv}\frac{\partial T_{pv}}{\partial t} = GA_{PV}\ \alpha_{pv}\ \tau_g\tau_{EVA} - P_{pv} + \frac{T_{EVA1} - T_{pv}}{R_{pv-EVA}} - \frac{T_{pv} - T_{EVA2}}{R_{pv-EVA}} \tag{5}$$

The hourly electric power output from the PV cell is given by Eq. (6) (Soliman and Hassan, 2019):

$$P_{pv} = A_{pv}G\ \alpha_{pv}\ \tau_g\tau_{EVA}\eta_{cell} \tag{6}$$

The instantaneous cell efficiency is given as a function of cell reference efficiency using Eq. (7) (Soliman et al., 2019):

$$\eta_{cell} = \eta_{ref}\left(1 - \beta_{pv}\left(T_{pv} - T_{ref}\right)\right) \tag{7}$$

The reference efficiency of PV cell η_{ref} is assumed 12.5% at a reference temperature T_{ref} of 298 K and β_{pv} is the temperature coefficient 0.0045 k^{-1} (Zhou et al., 2015). The optical properties of each layer of the PV (Glass, PV cell, EVA, Tedlar) are presented in Table 1.

3.1.4. Energy balance of lower EVA

$$M_{EVA}C_{EVA}\frac{\partial T_{EVA2}}{\partial t} = \frac{T_{pv} - T_{EVA2}}{R_{pv-EVA}} - \frac{T_{EVA2} - T_{TPT}}{R_{TPT-EVA}} \tag{8}$$

Table 1
Optical properties of PV panel for each layer (Zhou et al., 2015).

layers	Transmissivity	absorptivity	Emissivity
Glass	0.91	0.04	0.85
EVA	0.9	0.08	–
Silicon	–	0.9	–
TPT	–	–	0.9

3.1.5. Energy balance of TPT

$$M_{TPT}C_{TPT}\frac{\partial T_{TPT}}{\partial t} = \frac{T_{EVA2} - T_{TPT}}{R_{TPT-EVA}} - \frac{T_{TPT} - T_{ev}}{R_{ev-TPT}} \tag{9}$$

Thermal resistance between every two layers (R) is a conduction resistance (glass cover and upper EVA (R_{g-EVA}), EVA and silicon layer (R_{pv-EVA}), TPT and back EVA layer ($R_{TPT-EVA}$), and TPT layer and heat pipe evaporator wall (R_{ev-TPT}) and evaluated according to Eq. (10).

$$R = \frac{1}{2}\left[\left(\frac{x}{kA}\right)_{upper} + \left(\frac{x}{kA}\right)_{lower}\right] \tag{10}$$

where subscripts *upper*, *lower* represent two consecutive layers PV panel.

3.2. Thermal energy balance for HP

The heat balance of the heat pipe consists of heat balance for the evaporator and condenser. Heat pipe thermal resistance between evaporator and condenser $R_{con\text{-}ev}$ is the summation of several thermal resistances (Zohuri, 2011). It can be written as:

$$R_{con-ev} = R_{ev-ve} + R_{ev-wi} + R_{ev-wa} + R_{con-cond} + R_{con-wi} + R_{con-wa} \tag{11}$$

where evaporator section subscripts $ev-ve$, $ev-wi$, and $ev-wa$ denote the liquid evaporation process, the wicked metal, and the evaporator wall, the condenser subscripts $con-wi$, $con-wa$, and $con-cond$ denote the wicked metal, the condenser wall, and the vapor condensing process.

The thermal resistance terms in Eq. (11) are calculated as follows:

Thermal resistances R_{ev-wi}, R_{ev-wa}, R_{con-wa}, and R_{con-wi} are conduction resistances through a wall and calculated as follows:

$$R = \frac{x}{k\,A} \tag{12}$$

where x represents the wall thickness at the evaporator, condenser, and wick metal and k is the thermal conductivity of the heat pipe. A is the area of the evaporator and condenser section, $A = L\,D$, where L and D are the length and width of the evaporator and condenser sections,

respectively.

Wick thermal conductivity of the heat pipe for grooved wick is expressed as (Zohuri, 2011):

$$k_{wi}=\frac{wk_l\left[\left(0.185k_{wa}w_f+\delta k_l\right)+\left(\delta k_l k_{wa}w_f\right)\right]}{\left[\left(w+w_f\right)\left[\left(0.185k_{wa}w_f+\delta k_l\right)\right]\right]} \tag{13}$$

where w_f,w, and δ are the groove fin thickness, the groove width, and the depth of the groove, respectively.

Thermal resistances R_{ev-ve} and $R_{con-cond}$ are given by Eq. (14)(Hassan, 2014)

$$R=\frac{1}{A\,h} \tag{14}$$

where h is the film coefficient for evaporation and condensation of heat pipe fluid. The film coefficient for evaporation (h_{ev}) is given by Eq. (15), while The film condensation of acetone, h_{cond}, is given by Eq. (16) (Hassan and Harmand, 2015; Makki et al., 2016).

$$h_{ev}=\frac{k_l}{x_{wi}} \tag{15}$$

$$h_{cond}=1.13\left[\frac{g\,\sin(\theta)\rho_l(\rho_l-\rho_v)k_l^3h_{fg}}{\mu_L\Delta T_{vc}L_{con}}\right]^{0.25} \tag{16}$$

where θ is the tilt angle of the hybrid cooling system with a value of 30°.

3.2.1. Energy balance for HP evaporator section

The energy balance of the evaporator section is given by equation (17).

$$M_{ev}C_{HP}\frac{\partial T_{eva}}{\partial t}=\frac{T_{TPT}-T_{ev}}{R_{ev-TPT}}-\frac{T_{ev}-T_{con}}{R_{con-ev}}-\frac{T_{ev}-T_a}{R_{ev-a}} \tag{17}$$

The thermal resistance between the evaporator and ambient R_{ev-a}, given by equation (18).

$$R_{ev-a}=\frac{1}{h_{ev-a}A_{ev}} \tag{18}$$

$$h_{ev-a}=\frac{Nu_{ev-a}}{L_{ev}}k_a \tag{19}$$

The following empirical relation is used for calculating the Nusselt number Nu_{ev-a}. For laminar flow (Churchill and Chu, 1975),

$$Nu_{ev-a}=\left[0.825+\frac{0.387{Ra_a}^{1/6}}{\left(0.492\Big/Pr_a\right)\left(0.492\Big/Pr_a\right)\left(0.492/Pr_a\right)^{9/16}\Big]^{8/27}}\right]^2 \tag{20}$$

$$Ra_a=\frac{g\cos(\theta)B\,(T_{ev}-T_a)L_c^3}{\nu\,\alpha} \tag{21}$$

where k_a is the air thermal conductivity, ν is the kinematic viscosity of air, $\nu=\frac{\mu_a}{\rho_a}$, R_a is the Rayleigh number, $B=2/(T_{ev}+T_a)$ and α_a is the air thermal diffusivity, $\alpha_a=\frac{k_a}{\rho_a c_{p_a}}$.where Pr_a is Prendle number for air, $Pr_a=\frac{\mu_a\,c_{p_a}}{k_a}$. Table 2 shows the properties of air.

Table 2
Properties of ambient air and insulation.

Parameter	Value	Unit	Parameter	Value	Unit
Air specific heat, c_{p_a}	0.025	W·m^{-1} K^{-1}	Air dynamic viscosity, μ_a	1.8205x 10^{-5}	Pa.s
Air thermal conductivity, k_a	1006	J·kg^{-1} K^{-1}	Insulation thermal conductivity, k_{ins}	0.025	W·m^{-1} K^{-1}
Air density, ρ_a	1.2	kg·m^{-3}	Insulation thickness, x_{ins}	30	mm

3.2.2. Energy balance for HP condenser section

The energy balance of the condenser section is given by Eq. (22).

$$M_{con}C_{HP}\frac{\partial T_{con}}{\partial t}=\frac{T_{ev}-T_{con}}{R_{con-ev}}-\frac{T_{con}-T_{pcm}}{R_{con-pcm}} \tag{22}$$

the thermal resistance between the condenser wall and PCM is given by equation (30).

$$R_{con-pcm}=\frac{1}{2}\left(\left(\frac{x}{2kA}\right)_{con}+\left(\frac{x}{2kA}\right)_{pcm}\right) \tag{23}$$

3.3. Energy balance of PCM

The state of PCM (solid and liquid) depends on the applied temperature, and during phase change mode (from solid to liquid), the heat is transferred from the heated surface (HP condenser) to PCM. Both conduction and convection control heat transfer in PCM. However, natural convection can be ignored as PCM density changes are small (Soliman et al., 2021b). For simplicity, the convection mode is neglected in this model. For PCM, the total thickness of PCM is divided into layers, and the energy balance equation is applied for each layer. The thickness of each layer is referred to by x. Also, x represents the thickness of PCM from the condenser surface on one side (x equals half of the total thickness of PCM). The governing equations for PCM are presented by the following:

$$x_{pcm}\rho_{pcm}C_{pcm}\frac{\partial T_{pcm}}{\partial t}=\frac{1}{2}k\frac{\partial T}{\partial x} \tag{24}$$

The first layer that is directly in contact with the heat pipe condenser ($x=0$):

$$\frac{1}{2}k\frac{\partial T}{\partial x}=\frac{T_{con}-T_{pcm1}}{R_{con-pcm1}}-\frac{T_{pcm1}-T_{pcm2}}{R_{pcm}} \tag{25}$$

where $pcm1$ and $pcm2$ denote the first and the second layer from the condenser surface, respectively.

For the intermediate layers:

$$\frac{1}{2}k\frac{\partial T}{\partial x}=\frac{1}{2}k_{pcm}\frac{\Delta T_{pcm}}{\Delta x_{pcm}} \tag{26}$$

For the last layer (n):

$$\frac{1}{2}k\frac{\partial T}{\partial x}=\frac{T_{pcm,n-1}-T_{pcm,n}}{R_{pcm}}-\frac{T_{pcm,n}-T_a}{R_{pcm-a}} \tag{27}$$

The resistance between PCM and the ambient is given by Eq. (27)

$$R_{pcm-a}=\left(\frac{x}{2k}\right)_{pcm}+\left(\frac{x}{k}\right)_{ins}+\frac{1}{h_a} \tag{28}$$

An acrylic sheet is used as insulation for the PCM container, with properties and dimensions listed in Table 2.

3.3.1. Thermo-physical properties of PCM

In the following section, the properties of PCM (density, thermal conductivity, and specific heat) are presented (Abdelrazik et al., 2020) (Said and Hassan, 2021).

3.3.1.1. Density. As the density of PCM changes with the temperature, the density of PCM is a combination of liquid and solid density of PCM, as given in Eq. (29)(Said and Hassan, 2019).

$$\rho_{pcm}=\begin{cases}\rho_{pcm,s} & T_{pcm}\le T_{m1}\\ \lambda\rho_{pcm,l}+(1-\lambda)\rho_{pcm,s} & T_{m1}<T_{pcm}\le T_m\\ \rho_{pcm,l} & T_{pcm}>T_{m2}\end{cases} \tag{29}$$

the liquid fraction of the PCM (λ) is calculated as:

$$\lambda = \frac{T_{pcm} - T_{m1}}{T_{m2} - T_{m1}} \tag{30}$$

3.3.1.2. Thermal conductivity. The effective thermal conductivity of PCM is a combination of liquid and solid thermal conductivity of PCM as given in Eq. (31).

$$k_{pcm} = \begin{cases} k_{pcm,s} & T_{pcm} \le T_{m1} \\ \lambda k_{pcm,l} + (1-\lambda) k_{pcm,s} & T_{m1} < T_{pcm} \le T_m \\ k_{pcm,l} & T_{pcm} > T_{m2} \end{cases} \tag{31}$$

3.3.1.3. Specific heat capacity. Most of PCMs have a melting range rather than a melting point. Heat capacity will increase with the increasing melting range. The PCM specific heat in the case of temperature lower than the minimum phase change temperature is considered specific heat of solid ($C_{pcm,s}$), for higher than maximum phase change temperature is considered specific heat of liquid ($C_{pcm,l}$), and in between is taken a percentage. To account for latent heat, a piecewise function was used to describe and represent an effective specific heat capacity (Atkin and Farid, 2015).

$$C_{pcm} = \begin{cases} C_{pcm,s} & T_{pcm} \le T_{m1} \\ \dfrac{4LH - 4C_{pcm,s}\,(T_{m2}-T_{m1})}{(T_{m2}-T_{m1})^2}\left(T_{pcm} - T_{m1}\right) + c_{pcm,s} & T_{m1} < T_{pcm} \le T_m \\ \dfrac{4LH - 4C_{pcm,avg}(T_{m2}-T_{m1})}{(T_{m2}-T_{m1})^2}\left(T_{m2} - T_{pcm}\right) + c_{pcm,avg} & T_m < T_{pcm} \le T_{m2} \\ C_{pcm,l} & T_{pcm} > T_{m2} \end{cases} \tag{32}$$

where T_{m1}, T_{m2} are the start and endpoints of melting, while T_m the average melting temperatures of the PCM.

3.3.2. Thermal energy storage in PCM

Thermal efficiency of PCM (η_{th}) is defined as:

$$\eta_{th} = \frac{E_{th}}{GA_{pv}t} \tag{33}$$

Instantaneous thermal energy storage in PCM (E_{th}) is defined as:

$$E_{th} = \begin{cases} M_{pcm}c_{pcm,s}\left(T_{pcm,t} - T_{pcm,t_o}\right), & T_{pcm} \le T_{m1} \\ (1-\lambda)M_{pcm}c_{pcm,s}\left(T_{pcm,t} - T_{pcm,t_o}\right) + \Delta\lambda M_{pcm}LH, & T_{m1} < T_{pcm} \ge T_{m2} \\ (1-\lambda)M_{pcm}c_{pcm,s}\left(T_{pcm,t} - T_{pcm,t_o}\right) + \Delta\lambda M_{pcm}LH + \lambda M_{pcm}c_{pcm,l}\left(T_{pcm,t} - T_{pcm,m}\right), & T_{pcm} > T_{m2} \end{cases} \tag{34}$$

where $\Delta\lambda$, t, and t_o are the change in liquid fraction during an hour, instantaneous time, and initial time, respectively.

3.4. Exergy analysis

Exergy of the solar energy (Ex_{in}) is given as (Bayrak et al., 2017):

$$Ex_{in} = GA_{pv}\left(1 - \frac{4}{3}\left(\frac{T_a}{T_{sun}}\right) + \frac{1}{3}\left(\frac{T_a}{T_{sun}}\right)^4\right) \tag{35}$$

where T_{sun} is the solar radiation temperature, which is taken as 6000 K (Yousef and Hassan, 2020). Exergy stored in PCM is defined in Eq. (43) (Nie et al., 2019).

$$Ex_{pcm} = E_{th,sen,s}\left(1 - T_o ln\left(\frac{T_{pcm,t}}{T_{pcm,t_o}}\right)\right) + E_{th,latent}\left(1 - \frac{T_o}{T_m}\right) + E_{th,sen,l}\left(1 - T_o ln\left(\frac{T_{pcm,t}}{T_{pcm,m}}\right)\right) \tag{36}$$

where $E_{th,sen,s}$ and $E_{th,sen,s}$ are sensible heat stored in PCM for solid and liquid phase respectively, and $E_{th,latent}$ is the absorbed latent heat in PCM. T_o is denoted as the environment temperature, which is taken as the value of the initial ambient temperature.

The absorbing exergy efficiency of PCM (ζ_{pcm}) is defined in Eq. (37).

$$\zeta_{pcm} = \frac{Ex_{pcm}}{Ex_{in}} \tag{37}$$

The solar cell exergy efficiency (ζ_{pv}) is described as follows:

$$\zeta_{pv} = \frac{P_{pv}}{Ex_{in}} \tag{38}$$

3.5. Cost analysis

The intricacies of the cost evaluation may be shown using these stages (H. E. S. Fath, M. El-Samanoudy 2, 2003). A primary parameter in PV system cost analysis is the capital recovery factor, CRF, and it can be calculated as given in Eq. (39).

$$CRF = \frac{i(1+r)^n}{(1+r)^n - 1} \tag{39}$$

where n is the lifetime of the PV system, and r is the interest rate. The capital cost (CC) of the hybrid system is the cost of the system's main component (PV, HP, and PCM). The PV system's first annual cost (FAC) can be defined by Eq. (40).

$$FAC = CC\ x\ CRF \tag{40}$$

For PV system, the annual salvage value (ASV) is calculated as following:

$$ASV = S\ x\ SFF \tag{41}$$

For the PV system (PV panel, heat pipes, and PCM), the salvage value (S) is taken 7.5% of CC, while the sinking fund factor (SFF) can be obtained from Eq. (42)(Abo-Elfadl et al., 2021).

$$SFF = \frac{i}{(1+r)^n - 1} \tag{42}$$

The total annual cost (TAC) of the PV system is given as the following (Yousef and Hassan, 2020):

$$TAC = AMC + FAC - ASV \tag{43}$$

Annual maintenance cost (AMC) is taken as 15% of FAC.

The price of the kilowatt-hour of electricity (P_e), in ($/kWh), can be calculated by (Yousef et al., 2022):

$$P_e = \frac{TAC}{E_{annual}} \tag{44}$$

where E_{annual} is the annual electric power output. The payback period (PB) is calculated by (Adun et al., 2022).

$$PB = \frac{CC}{AS} \tag{45}$$

where annual net cost savings (AS) represents the cost savings realized after all operating expenses are fulfilled.

$$AS = \text{Annual savings} - \text{AMC} \tag{46}$$

4. Model solution and validation

The previous transient governing equations are solved together and programmed inside MATLAB software. These systems of equations are solved numerically using a homemade program based on the fourth-order Runge-Kutta method. MATLAB 2021b software is used to perform the solution and analyze the data. In the following step, the validation of the numerical solution is stated by using both PV model results and PV-HP model results from the literature.

4.1. PV model validation

The theoretical results of the PV model are validated by using the results of (Yousef et al., 2016; Zhou et al., 2015). Firstly, the current model results for the conventional PV module are validated with the numerical results of uncooled PV cells (Zhou et al., 2015). Fig. 2 illustrates a comparison between the results of the present model and the numerical results of the Zhou et al. model for the PV panel cooled by natural convection and radiation. Fig. 2a and b, respectively, show the temperature difference between the solar cell and ambient temperature variation with varying ambient temperatures and solar radiation. According to Fig. 2a, the maximum, minimum, and average differences between the current results and the numerical results of the Zhou et al. model are 6%, 0.27%, and 2.4%, respectively. These values are 6%, 0.1%, and 2.9%, respectively for Fig. 2b. Secondly, the experimental results of (Yousef et al., 2016) are used to validate the variation of the PV cell temperature of the current model, which is coded by both radiation and natural convection. Fig. 3 reveals the variation of conventional PV cell temperature during the daytime of the present model and experimental data obtained by Yousef et al. The comparison shows a homogenous agreement between the current results and that of Yousef et al.

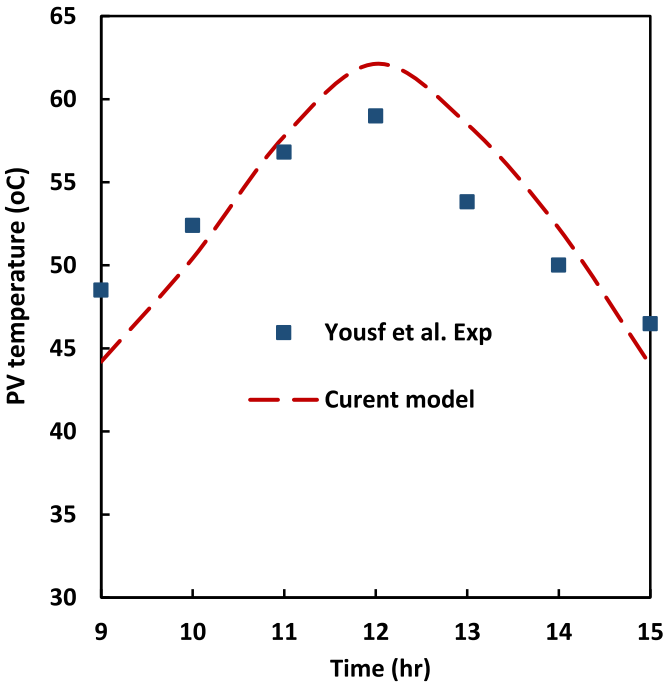
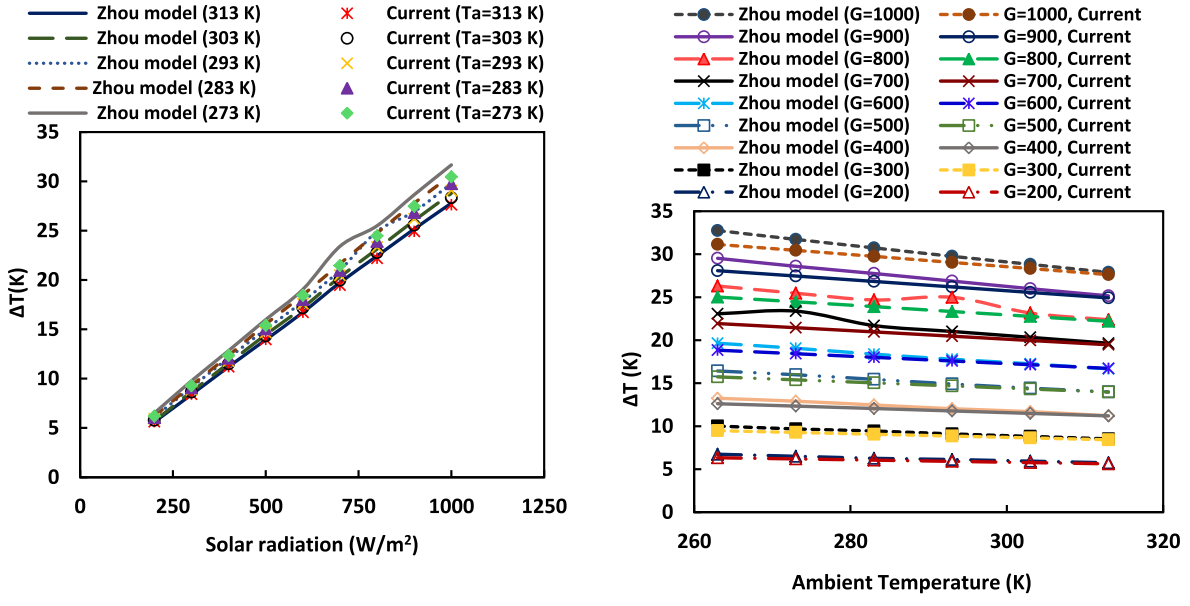


Fig. 3. Variation of PV temperature with time for the current model and Yousef et al. results (Yousef et al., 2016).

experiment with maximum, minimum, and average error about 8.85%, 1.6%, and 5.4%, respectively.

4.2. PV-HP model validation

Another validation, where all experimental parameters are considered, is done by corroborating the current PV and heat pipe system with the experimental results of (Soliman and Hassan, 2020). This experiment used heat pipes to absorb and transfer excess heat from the PV panel. Simultaneously, a heat sink (HS) was used to cool the condenser section of the heat pipe. Additionally, A solar simulator was used to



(a) Ambient temperature (b) Solar radiation

Fig. 2. Comparison between the current model and Zhou et al. (Zhou et al., 2015) model results at different (a) ambient temperature and (b) solar radiation intensities.

Table 3
Properties and dimensions of the flat heat pipe (1070 Aluminum, 2022; Chen and Chou, 2014; Han et al., 2020; Zohuri, 2011).

Parameter	Value	Unit	Parameter	Value	Unit
Total length, L	500	mm	Heat pipe density , ρ_{HP}	2700	kg·m^{-3}
Width, D_{ev} = D_{con}	250	mm	Wall thermal conductivity, k_{wa}	230	W·m^{-1} K^{-1}
Condenser length, L_{con}	200	mm	Heat pipe specific heat capacity , C_{HP}	900	J·kg^{-1} K^{-1}
Evaporator length, L_{ev}	250	mm			
Wall thickness, x_{wa}	0.8	mm	The porosity of wick, φ_{wi}	50	%
Wick thickness, x_{wi}	0.4	mm	Acetone thermal conductivity, k_l	0.168	W·m^{-1} K^{-1}
Groove fin thickness, w_f	0.2	mm	Acetone liquid density	744	kg·m^{-3}
Groove width, w	0.2	mm	Acetone vapor density	2.37	kg·m^{-3}
Groove, δ	0.4	mm	Viscosity of acetone, μ_L	0.226	Pa.s
Vaporization heat of -acetone, h_{fg}	518	kJ·kg^{-3}	The temperature difference between water -and heat pipe wall, ΔT_{vc}	273	K

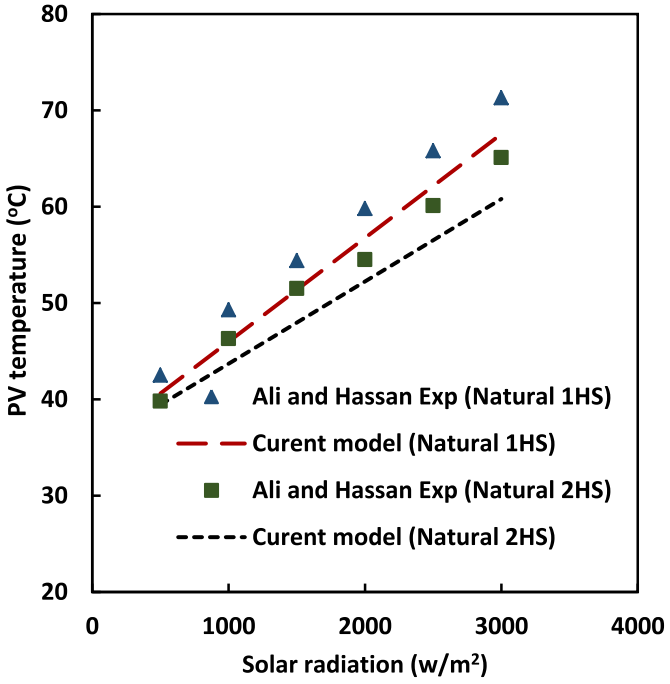


Fig. 4. Variation of PV temperature with incident radiation for Soliman and Hassan (Soliman and Hassan, 2020) and the current model.

simulate solar radiation with values varying from 500 W/m^2 to 3000 W/m^2. In PV-HP validation, the cell temperature is cooled by a flat heat pipe, with the dimensions and properties listed in Table 3, and PCM is replaced with a heat sink, where both a single heat sink (1HS) and double heat sink (2HS) are used. Fig. 4 illustrates PV cell temperature variation with simulated solar radiation intensity for natural convection cooling of heat pipe condenser. An agreement is shown from the comparison, with maximum, minimum, and average errors of about 6.8%, 0.9%, and 5.2%, respectively.

5. Results and discussion

The paper results are divided into five subsections. The first subsection demonstrates the impact of using the integrated HP-PCM cooling system on the PV cell performance. This subsection investigates the performance of the PV panel (cell temperature) under varying PCM thickness (from 10 mm to best thickness) for different PCM types and PCM amount is changed by changing its height. The behaviour of PCM is depicted in the second subsection, including the liquid fraction and the temperature evolution in PCM for different types and amounts of PCM under both weather conditions. The proposed system's impact on instantaneous PV electric and thermal efficiencies is presented along with PV electric energy output, PCM thermal energy storage, the average electric and thermal efficiencies, and total thermal energy for each PCM type in the third subsection. The fourth subsection illustrates the performance of exergy analysis for the PV system and PCM. Finally, the fifth subsection reports on the economic analysis for the proposed hybrid system compared with PV only.

Table 4
PV panel properties and dimensions for each layer (Makki et al., 2016; Soliman and Hassan, 2019).

layers	Thickness (mm)	Thermal conductivity (W·m^{-1} K^{-1})	Specific heat capacity (J·kg^{-1} K^{-1})	Density (kg·m^{-3})
Glass	3.2	2	500	2450
EVA	0.5	0.311	2090	960
Silicon	0.2	148	677	2330
TPT	0.3	0.15	1250	1200

The dimensions and properties of the used PV cell in the current results are listed in Table 4. The performance assessment for the PV/HP-PCM arrangements is computed under two different climate conditions of summer and winter days. The solar radiation on the PV model and ambient temperature are measured on the third of August, for summer and the first of January, for winter conditions in Assiut city, Egypt. The quantity of the PCM is expressed by PCM thickness, where its length equals the condenser length, and its width equals heat pipe and PV width; both the length and width of PCM are assumed fixed in all cases.

In the current results, conventional PV means that the PV is cooled naturally by the air without using HP or PCM. Initially, the PV panel, heat pipe (evaporator and condenser section), and PCM are assumed at an ambient temperature of 31 °C for summer and 13.7 °C for winter operation. In this study, the initial temperature of PCM is considered at a temperature higher than the night temperature. As depicted in Fig. 1, it is assumed that the container of the PCM is insulated; also, the lower surface of the evaporator is subjected to convection heat transfer. The upper boundary conditions of the PV are the same as that of the PV in the PV/HP-PCM system. Similarly, the lower boundary conditions of the PV are the same as that of the lower evaporator heat pipe condition (Natural convection). Boundary conditions are shown in Fig. 1.

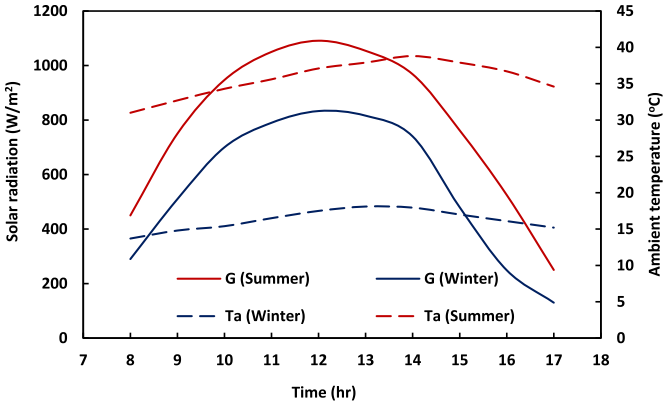


Fig. 5. Variation of solar radiation intensity and ambient temperature with time.

Table 5
Thermo-physical properties of used phase change materials, summer case (Rubitherm, 2022).

Properties		RT43	RT35HC	SP31
Melting range, °C, Tm		37–43	34–36	31–33
Latent heat, J·kg^{-1}, LH		165000	240000	210000
Thermal conductivity, W·m^{-1} K^{-1}, k_{pcm}	solid	0.2	0.2	0.5
	liquid	0.2	0.2	0.5
Density, kg·m^{-3}, ρ_{pcm}	solid	880	880	1350
	liquid	760	770	1300
Specific heat capacity, J·kg^{-1} K^{-1}, c_{pcm}	solid	2000	2000	2000
	liquid	2000	2000	2000

Table 6
Thermo-physical properties of used phase change materials, winter case (Rubitherm, 2022).

Property state		RT21	RT18HC	SP15-gel
Melting range, °C, Tm		18–23	17–19	15–17
Latent heat, J·kg^{-1}, LH		155000	260000	160000
Thermal conductivity, W·m^{-1} K^{-1}, k_{pcm}	solid	0.2	0.2	0.6
	liquid	0.2	0.2	0.6
Density, kg·m^{-3}, ρ_{pcm}	solid	880	880	1400
	liquid	770	770	1350
Specific heat capacity, J·kg^{-1} K^{-1}, c_{pcm}	solid	2000	2000	2000
	liquid	2000	2000	2000

Fig. 5 illustrates the intensity of the solar radiation and ambient temperature variation with time. Fig. 5 reveals that the intensity of solar radiation is sored with time until reaching its peak of 1091 W/m^2 for summer and 833 W/m^2 for winter at solar noon; however, it drops by the end of the day. Similarly, ambient temperatures experienced similar solar radiation behaviour; however, the summer day's peak temperature is 38.8 °C at 14:00 a.m., and the winter day's peak temperature is 18.1 °C at 13:00 a.m., as shown in Fig. 5. Wind speed, in this study, is taken as an average value of 1.5 m/s for summer conditions and 2.5 m/s for winter conditions.

Thermophysical properties of PCMs used in summer and winter are listed in Tables 5 and 6, respectively. This study selected three PCM types for summer and winter simulation. The selection is based on the melting range of PCM. The melting range should be close to the operating temperature of the PV panel, and the solidification point should be slightly higher than the ambient night temperature.

5.1. Impact of cooling system on PV temperature

Fig. 6a and b shows the influence of using the hybrid HP-PCM cooling system for different PCM types and amounts on the hourly operating temperature of the PV panel for both summer and winter conditions, respectively. The operating temperature of the PV panel rises with increasing the intensity of solar radiation and ambient temperature, which peaked at noon. However, as the solar radiation and ambient temperature fall in the afternoon, the panel's temperature declines as well. For all PCMs used, there is a considerable reduction in the operating temperature of the PV/HP-PCM system compared with natural PV cooling. For a small amount of PCM, 10 mm on one side of the condenser section, the cell temperature of the hybrid system is lower than the conventional PV; nonetheless, when PCM completely melts, the hybrid system's cell temperature rises higher than the temperature of PV only. During the melting process, a large amount of heat is absorbed by PCM as latent heat, and after melting, PCM is heated sensibly. As a result, less heat is absorbed from the PV panel. In the afternoon, less heat is generated from the cell. Moreover, as the energy storage in PCM is higher than the rate of heat generation from the PV panel, PCM goes into a cooling process. Consequently, the cooling rate of the PV/HP-PCM panel was significantly reduced (Luo et al., 2017). By increasing the PCM amount, the period of both thermal regulation and PCM melting increases.

The optimum amount of PCM is required for complete melting by the end of the day. Furthermore, it depends on PCM properties (melting range and latent heat of fusion). For example, a small amount of PCM is required if the PCM has a high latent heat of fusion, as demonstrated in Fig. 6a (case RT35HC) and 6b (case RT18HC). On the other hand, a large amount is required for a low melting point, as depicted in Fig. 6a (case SP31) and 6b (case SP15-gel). The results show that the required PCM quantity that avoids reverse heating for the PV in summer is 30, 25, and 30 mm for RT42, RT35HC, and SP31, respectively. These values for the winter case are 25, 20, and 25 mm for RT21, RT18HC, and SP15-gel, respectively.

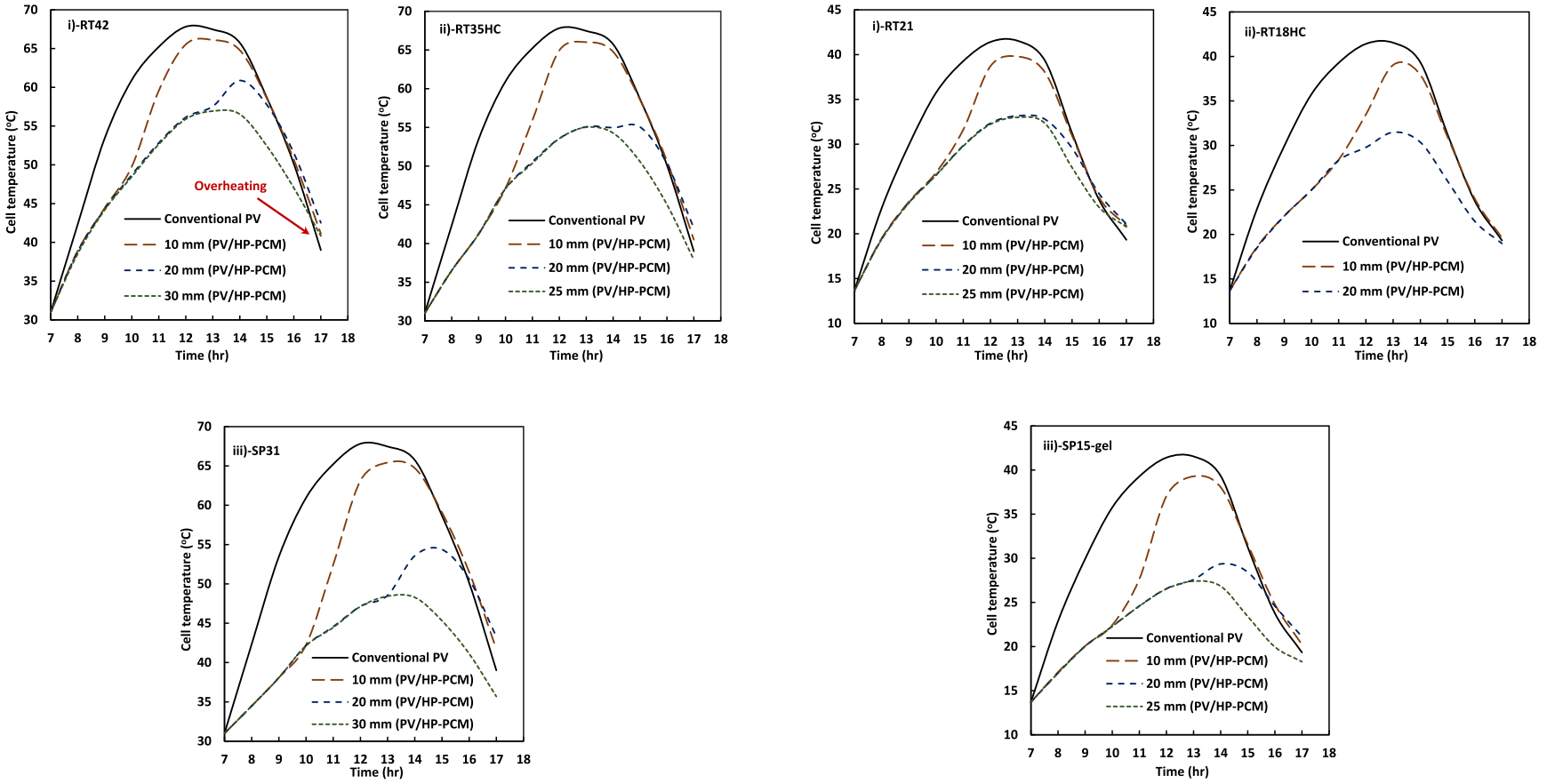
It is clear from the results that decreasing the melting temperature of the PCM reduces the PV maximum temperature, but this requires more time to solidify the PCM during nighttime. Raising the PCM quantity reduces the PV temperature, especially at higher solar intensity. Additionally, during the melting of the PCM, it is clear that the PCM amount has no sensible impact on the PV temperature. In winter, the PV temperature behaviour is approximately similar to summer but with a lower value due to smaller solar intensity.

The results show that in the summer, RT42 and RT35HC reduce the maximum operating temperature of the PV/HP-PCM panel by 11.9 °C and 14.2 °C, respectively, for 30 mm and 25 mm thickness, respectively. The maximum reduction in PV temperature accounts for SP31 with 20.6 °C lower than conventional PV panel at PCM thickness of 25–30 mm. In the winter, the maximum operating temperature of the conventional PV panel is 41.5 °C compared with 33 °C for RT21 and 31.4 °C for RT18HC. However, SP15-gel achieved the maximum reduction of the PV temperature of 14.1 °C with a maximum operating temperature of 27.4 °C of 25 mm thickness. The hybrid cooling system is better than the naturally cooled PV panel for all PCM types. Furthermore, SP31 and SP15-gel have better thermal temperature control than other PCM types.

5.2. PCM thermal behavior

5.2.1. Liquid fraction

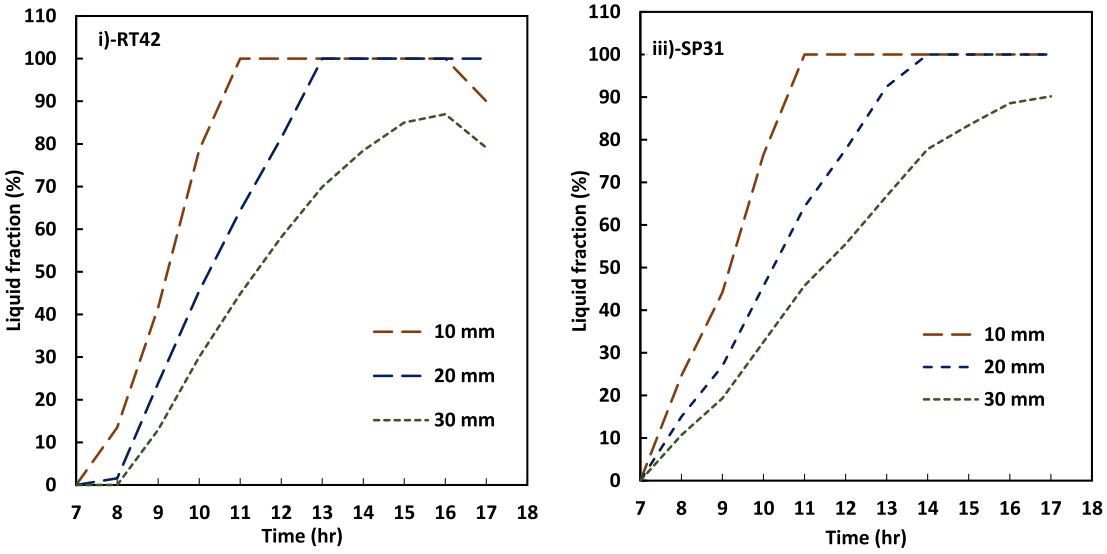
Fig. 7a and b shows the variations of the percentage of the liquid fraction of the PCM with time for different amounts of PCM for summer and winter conditions, respectively. All graphs have an increase in the liquid fraction with time. Moreover, the complete melting of the PCM is faster for the small amount of PCM compared with the larger amounts. For example, for a 10 mm thickness of PCM, the complete melting is achieved before midday for all PCM types. However, the complete melting for small thickness for RT18HC is achieved at noon because RT18HC has a relatively high melting point and latent heat of fusion compared with other winter types. By increasing PCM amount, PCM takes a long time for complete melting; this long time is desirable for thermal management of the PV panel, as discussed previously. Results from Fig. 7a for RT42 in summer and 7b for RT18HC in winter depict that PCM starts to solidify (the melting fraction decreases) by the end of the day. This solidification occurs because RT42 and RT18HC have a relatively high melting point; as time advances, the rate of heat transfer to PCM decreases due to decreasing the rate of heat generation from the PV cell. As a result, heat transfers from hot PCM to relative cold PV by the end of the day, solidifying part of PCM (Reconduction of heat to the PV). In order to avoid the re-conduction of heat, the PCM melting temperature must be decreased as much as possible, and its latent heat and quantity must be increased. However, decreasing the PCM melting temperature requires a longer time to solidify the PCM during the nighttime. Re-solidification of RT21 occurs before RT18HC because the melting point of RT21 is high compared with RT18HC. From these results, decreasing PCM melting point attenuates the re-solidification process. The liquid fraction behavior of RT35HC and SP15-gel is similar to SP31.



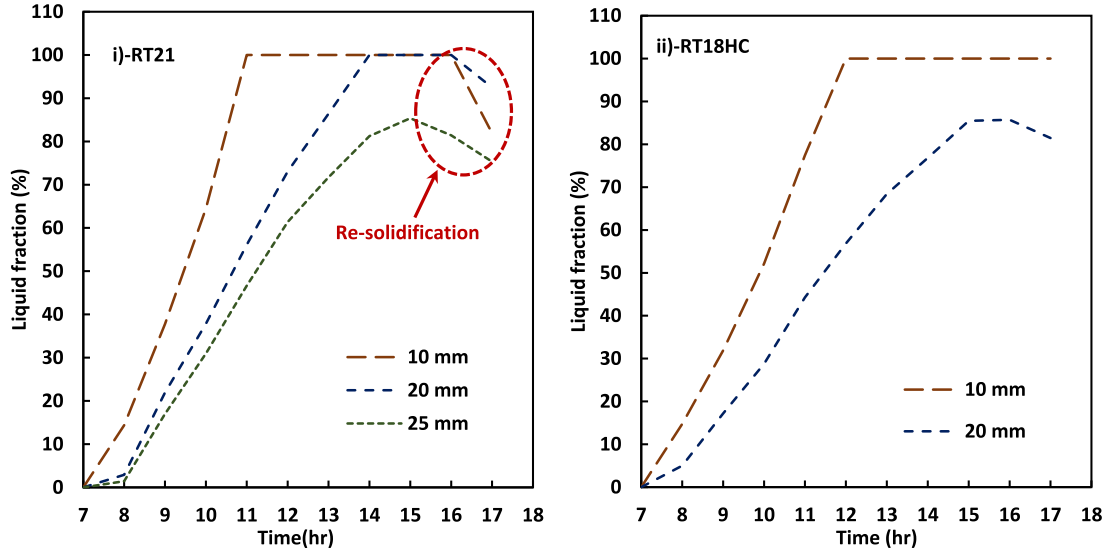
(a) Summer (PCM best thickness of RT42, RT35HC, and SP31 are 30, 25, 30 mm, respectively)

(b) Winter (PCM best thickness of RT21, RT18HC, and SP15-gel are 25, 20, 25 mm, respectively)

Fig. 6. Hourly variation of cell temperature for PV/HP-PCM system for different PCM types compared with natural PV cooling in (a) summer and (b) winter.



(a) Summer



(b) Winter

Fig. 7. Liquid fraction variation with time for each PCM type for (a) summer and (b) winter.

5.2.2. PCM temperature evolution

Fig. 8a and b illustrate the variation of temperature evolution of PCM with time for summer and winter cases, respectively. The best PCM thickness temperature distribution is shown. Depending on the melting range, PCM is initially heated sensibly till it reaches the melting point, and a small amount of heat is absorbed from the PV cell, as depicted clearly in Fig. 8a for RT42 and RT35HC and 8b for RT18HC and SP15-gel. This sensible heat represents vertical-inclined lines from the initial. After that, the PCM achieves phase change due to the energy absorbed with a relatively small change in PCM temperature (PCM melting range). After complete PCM melting, its temperature sensibly increases. The PCM layers near the condenser surface are quickly heated, while the layers further away from the condenser surface take longer to heat up. The heat resistance of PCM layers increases as the PCM layer thickness increases from the condenser section. The results indicate that PCM temperatures increase with time as solar radiation increases for all PCM types studied. Following that, as the intensity of solar radiation decreases, the temperature of the PCM layers near the condenser section decreases due to the decrease in heat transport from the PV to the PCM. However, the temperature of the top PCM layers continues to rise because the reduction in solar intensity has little effect on them, and they still receive heat from liquid-hotter layers.

The first layer of PCM (for example, 5 mm thickness above the condenser section) is heated rapidly as heat transfers by conduction through a small layer thickness. PCM temperature decreases with the increasing thickness of PCM from the condenser section because the thermal resistance of PCM layers increases. The zone of the constant PCM temperature during the melting period is for the PCM far from the condenser because of the lower heat transfer rate inside these zones compared with the zone near the condenser. The results show that the comportment of the different PCMs in winter is similar to that in summer, but the PCM temperature is lower. For example, the maximum PCM temperature at 5 mm thickness in summer is about 51.8 °C, 48.5 °C, and 41.3 °C, in the case of RT42, RT35HC, and SP31, respectively, while this value is nearly 25.9 °C and 21.5 °C in the case of RT18HC and SP15-gel, respectively. It is worth mentioning that PCM temperature is strongly

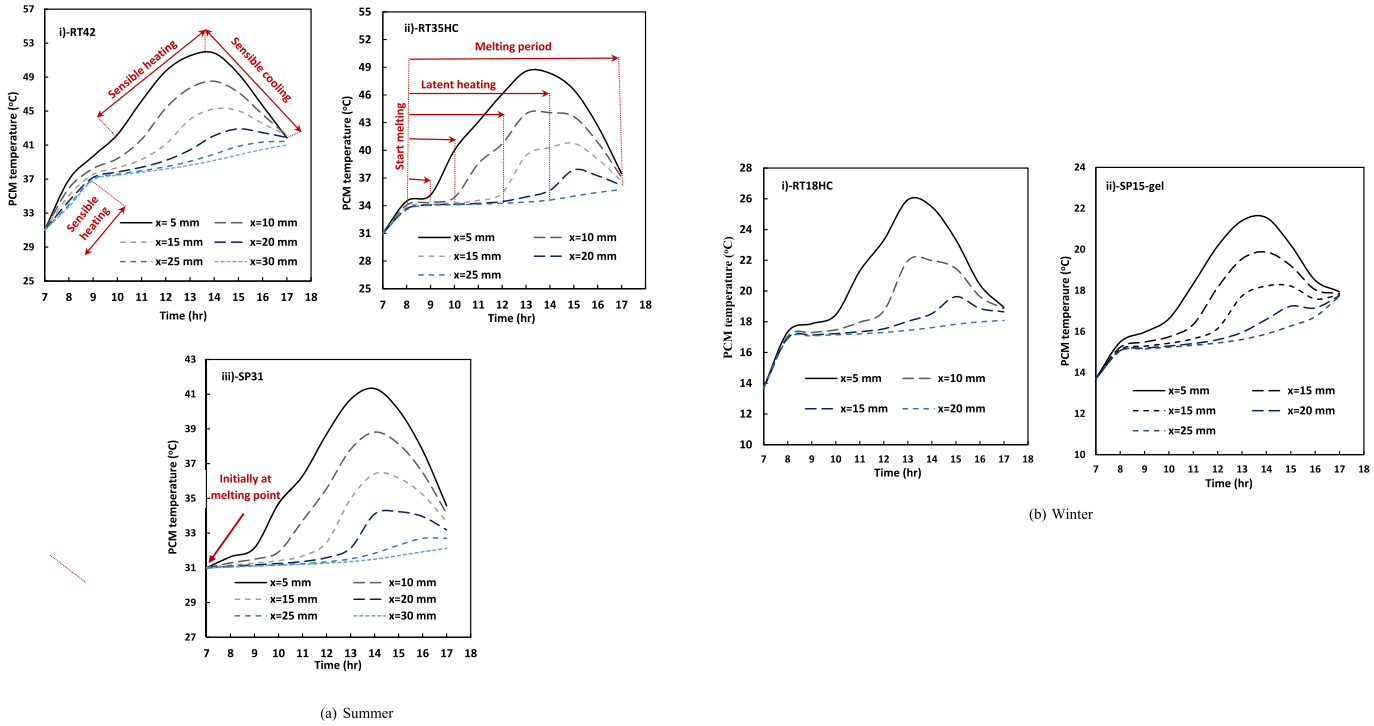


Fig. 8. Evolution of PCM temperature variation with time for best PCM thicknesses in (a) summer and (b) winter.

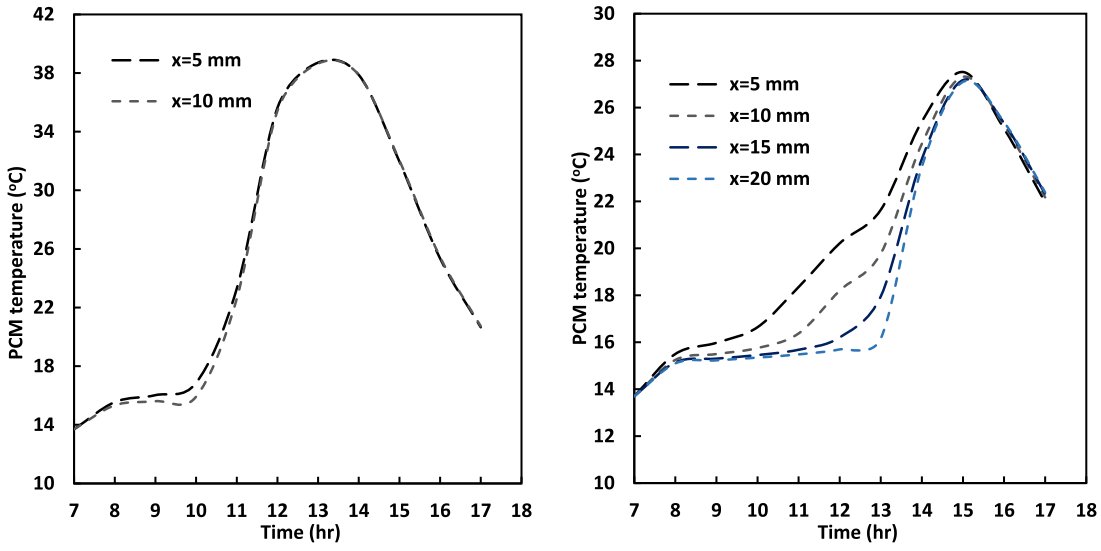


Fig. 9. Impact of PCM amount on the temperature distribution inside PCM layers in case of SP15-gel type.

affected by its thickness due to that most PCMs have a melting range rather than a melting point. As a result, PCM whole thickness is divided into layers of small thickness because each layer has a different temperature due to different melting times (Abdulmunem et al., 2021; Emam et al., 2019). They measured PCM temperature by dividing PCM whole thickness into layers (positions of thermal couples). Based on these studies, the temperature of the layer in the current study is taken as the average temperature of the layer. This study is considered for 0D analysis of the PCM, and further investigation could be performed to study PCM in more than one dimension.

Fig. 9 shows the effect of PCM amount on the temperature evolution of SP15-gel for total PCM thickness 10 and 20 mm. When a small amount of PCM is used (10 mm), complete melting for all PCM layers is achieved quickly, before noon, as shown in Fig. 9a. Before the complete melting period, the layer of the PCM adjacent to the condenser melts faster, and the melting rate decreases with departing from the condenser. Accordingly, the temperature of the PCM changes with changing the distance from the condenser. After complete melting, the PCM layers' temperature increases due to sensible heating. Despite the rise of solar radiation during this period, the heat transfer from the PV decreases slightly compared to the case before the complete melting of the PCM. Hence, the increase in the temperature of layers near the condenser will be lower due to the decrease in the heat transfer rate from the PV side and the heat loss to the upper layers. As discussed earlier, backflow heat

reduces the PCM temperature and increases the PV panel temperature. That results in a decrease in the PCM temperature having the same value. In this case, the PCM temperature approaches a lumped system, especially for lower total PCM thickness. The lumped approach occurs because the layers near the condenser lose their heat to the PV direction, but at the same time, it gains heat from the farther layer. However, the upstream layers lose heat at a lower rate.

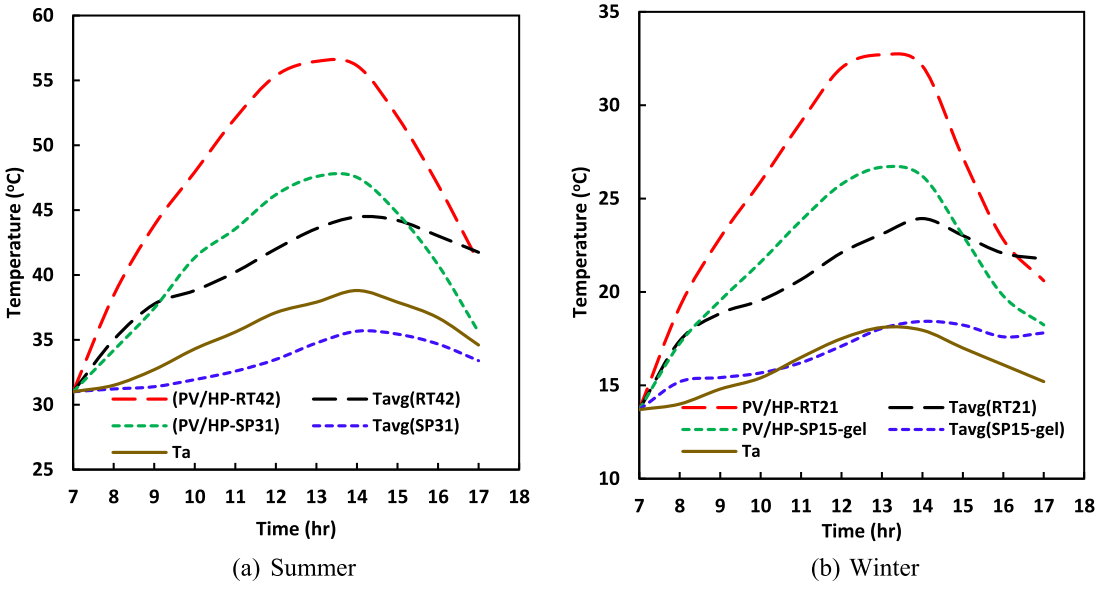
Increasing SP15-gel total thickness to 20 mm, the complete melting takes longer than smaller thickness, as shown in Fig. 9b. It is clear from these figures that within the melting period, the PCM temperature increases despite decreasing the solar intensity afternoon, which lowers the PV temperature, as shown early. There is a difference in temperature during the melting period for different distances from the condenser section. Generally, the temperature of the position near the condenser has a higher value than that far from the condenser. Moreover, afternoon, at any position inside the PCM reaches complete melting, its temperature decreases again. There is a clear difference between layers' temperatures during the melting process, about 5.5 °C at 13:00. The temperature difference in PCM decreases gradually due to two reasons. Firstly, more heat is transferred from hot-liquid layers to less hot-solid layers; secondly, the heat transfer rate from the PV system to PCM layers declines with decreasing solar intensity. Since complete melting for 20 mm thickness takes more time, better thermal regulation is achieved until this time. Based on the previous results, it is recommended to choose the PCM quantity that its melting process lasts until the end of the PV working day. However, this requires a more extended period of solidification during nighttime, which must also be considered.

Fig. 10a and b shows the variation of ambient air temperature, PCM average temperature, and panel backside temperature of PV/HP-PCM for two PCM types in summer and winter, respectively. It is observed that the profiles of average PCM temperature and panel backside temperature of the hybrid cooling system are similar to the ambient temperature profile. Fig. 10a shows that the temperature of the panel backside is higher than the ambient temperature and average PCM temperature using RT42 by 18.5 °C and 12.9 °C at 13:00, respectively. These values for SP31 are 9.6 °C and 12.8 °C. from these results, using SP31 (low melting point compared with other summer PCM) brings the cell temperature close to ambient temperature. Also, it is noticed that the average temperature of SP31 is lower than ambient temperature. This temperature difference is because SP31has a melting range (31 °C–33 °C) lower than ambient temperature. Consequently, the PCM container is insulated to prevent heat flow from the ambient. In contrast, RT42 has an average temperature higher than the ambient as it has a high melting range (37 °C–42 °C). As discussed previously, the temperature of PCM is higher than the panel backside in the case of RT42 and RT21due to backflow heat. In winter, the behavior of two PCM is similar to summer's types.

5.3. PV electric performance

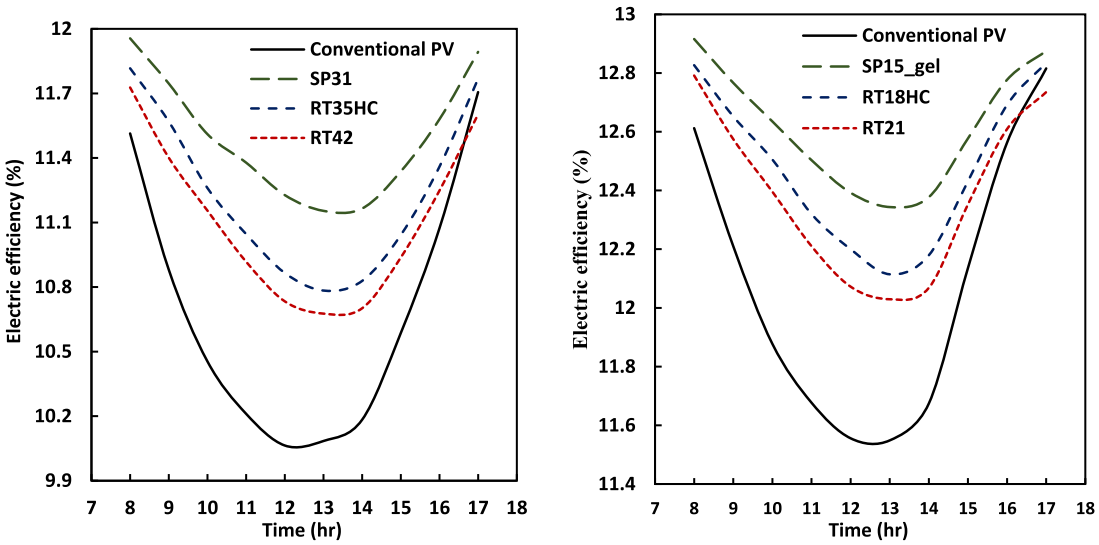
5.3.1. Electric efficiency of PV

The electric efficiency of the PV cell depends on the cell's operating temperature, as stated in Eq. (7). As mentioned in the introduction section, the high operating temperature of the solar cells reduces the efficiency of the panels. Fig. 11 shows the variation of electric efficiency of the PV cell for the proposed cooling system compared with the conventional PV panel for summer and winter conditions at the best PCM thickness. The graphs show that the electric efficiency decreases with time until noontime because the cell temperature increases with time until midday. After that, the electric efficiency rises again as the temperature of the PV panel drops, taking the inverse trend of the solar intensity and ambient temperature. Electrical efficiency is affected by several parameters, such as solar radiation, cell temperature, and ambient temperature, as can be seen from the data in Fig. 11, which shows that summer efficiency is lower than winter efficiency. The electric efficiency of the PV/HP-PCM system is higher than the conventional one due to the reduction in cell operating temperature. The maximum enhancement in the electric efficiency for the hybrid PV panel using PCM types RT42 and RT35HC is 6.6% and 7.9%, respectively, compared to convention PV. SP31 achieved the maximum enhancement with 11.5% (1.2% increase) because SP31 achieves the maximum temperature reduction compared with other PCM types. Winter's electricity efficiency profile mirrored summer's profile. The maximum improvement for winter types RT21, RT18HC, and SP15-gel is 4.4%, 5.5%, and 7.2% (0.9% increase), respectively. Additionally, the SP31 system's electrical efficiency is more consistent; the fluctuation in value between the lowest and highest efficiency is roughly 0.79%, compared to 1.64%, 1.05%, and 1.03% for the PV, RT42, and RT35HC systems, respectively. The conventional panel's electrical efficiency exceeded that of the RT42 and RT21 systems by the end of the thermal regulation period, as the temperature of the hybrid system grew more than that of the PV system alone, as previously described. If the present results are compared with that obtained by (Yousef et al., 2022). It is found that they obtained a maximum increase in the efficiency using a PCM attached to the PV is



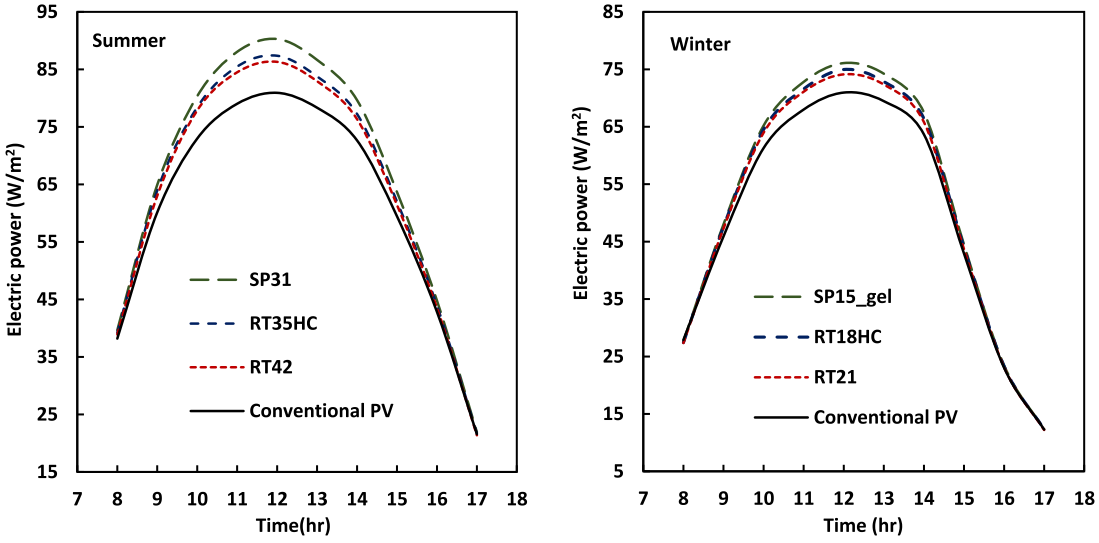
(a) Summer (b) Winter

Fig. 10. The variation of temperatures of ambient air (Ta), PV/HP-PCM, and the average temperature of PCM with time in (a) summer and (b) winter.



(a) Summer (b) Winter

Fig. 11. Variation of electrical efficiency of PV cell for different PCM types during daytime compared with the conventional PV panel for (a) summer and (b) winter.



(a) Summer (b) Winter

Fig. 12. Variation of electrical power of PV/HP-PCM system for different PCM types during daytime compared with the conventional PV panel for (a) summer and (b) winter.

about 1.1% and 1.4% in winter and summer, respectively. Hence, the current increase of the PV efficiency using PCM-Heat pipe system is closed to the previous results with a maximum difference of about 0.2%. However, the present system stores the heat inside the PCM far a way from the PV panel.

5.3.2. PV electric power

Fig. 12 shows the variation of electric power of the PV panel integrated with the hybrid cooling system and conventional PV panel for both summer and winter conditions using different PCM types. According to Eq. (6), electric power output depends on the intensity of solar radiation, and it is clear that electric power goes up with time with increasing solar intensity until noontime. After that, the electric power for the hybrid PV and PV declines by falling solar radiation. Moreover, electric power depends also on electric efficiency, which functions on cell temperature. The maximum electric power is achieved at noontime for summer and winter conditions. The electrical power in summer is greater than in winter, despite the efficiency of winter being greater than in summer, and it has the smallest value at noontime. From the previous discussion, solar intensity is the dominant factor in the PV output electrical power. As clearly depicted in Fig. 12, in summer, the electrical power in descending order for different PCM types is for SP31, RT35HC, and RT42, and in winter is for SP15-gel, RT18HC, and RT21. This order means that decreasing the melting point of the PCM increases the electric power. Also, the maximum instantaneous electric power is high for the hybrid PV panel for all studied PCM types. Among the different PCMs types, the ultimate upgrade in electric power accounts for SP31 in summer and SP15-gel in winter with 11.5% and 7.2%, respectively. Their average electric power improvement is 7.8% and 4.3%, respectively. However, other summer PCM types, TR35HC and RT42, have maximum enhancement of 7.9% and 6.6%, respectively, while their upgrade of the average electric power is 5.3% and 4.2%, respectively, compared with conventional PV. On the other hand, for RT18HC, the maximum and average improvements are 5.6% and 3.1%, and for RT21

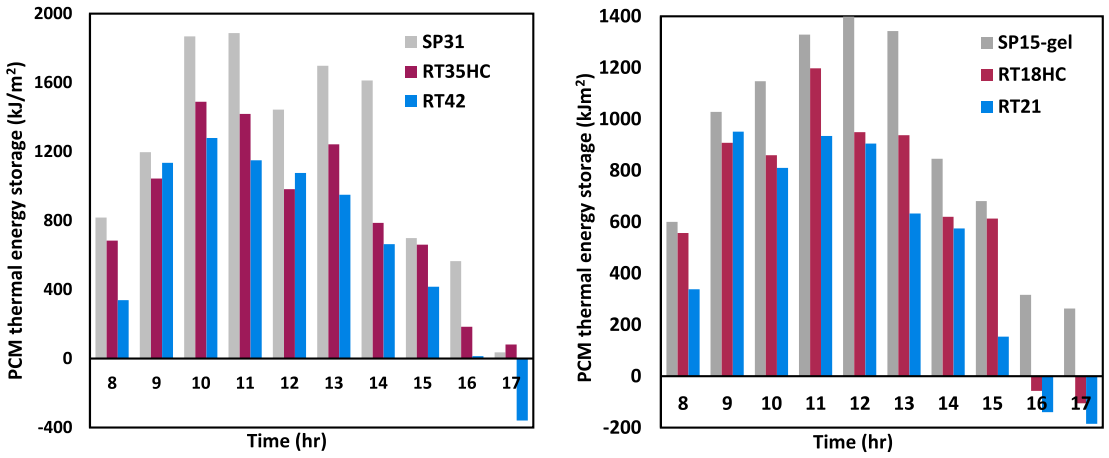


Fig. 13. Hourly stored thermal energy by PCMs in (a) summer and (b) winter.

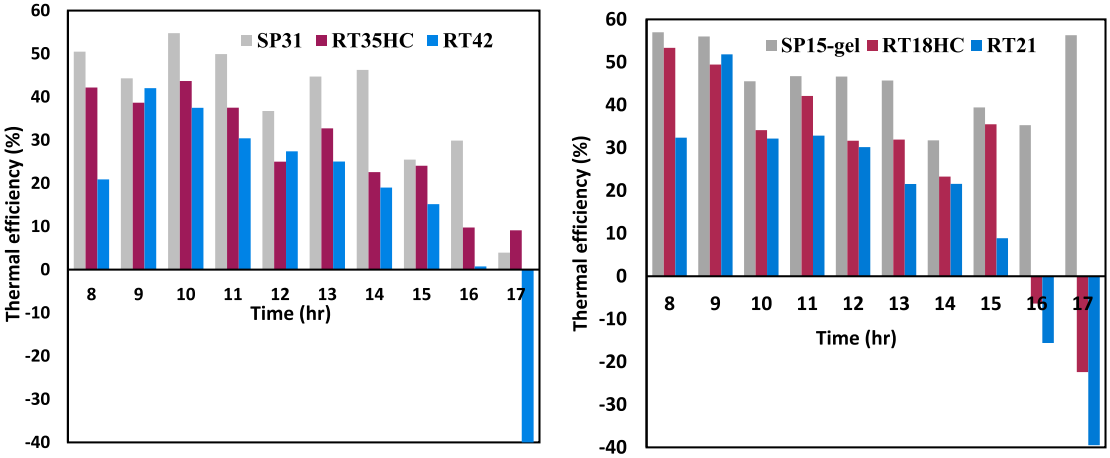
are 4.4% and 2.3%, respectively, for winter compared to conventional PV. Furthermore, in contrast to the summer results, the discussed data indicated that employing PCM was less advantageous for electric power enhancement in winter when solar radiation and ambient temperature are low.

5.4. PV/HP-PCM system performance

Fig. 13a and b shows the hourly thermal energy absorbed from the hybrid PV system and storage in PCM per square meter of PV area (thermal energy storage of the system) for summer and winter cases, respectively. Axiomatically, the maximum hourly thermal energy storage in summer conditions is higher than in winter because the solar radiation intensity in summer is higher than in winter; consequently, more generated heat from the PV panel. Thermal energy storage in PCM increases with time until the complete melting of the PCM, and then it drops again. Heat is initially absorbed sensibly; then, the stored thermal energy rises due to melting the PCM and rising the solar radiation until noon. By the time a large portion of PCM is converted to liquid phase, the temperature of PCM rises, and solar radiation plummets; as a result, the heat transfer rate to the PCM starts to decline, as explained earlier, thermal energy storage decreases. Due to their proper conditions (relatively high latent heat of fusion and low melting point), SP 31 and SP15-gel have the maximum energy storage ability for summer and winter conditions compared with other PCM types.

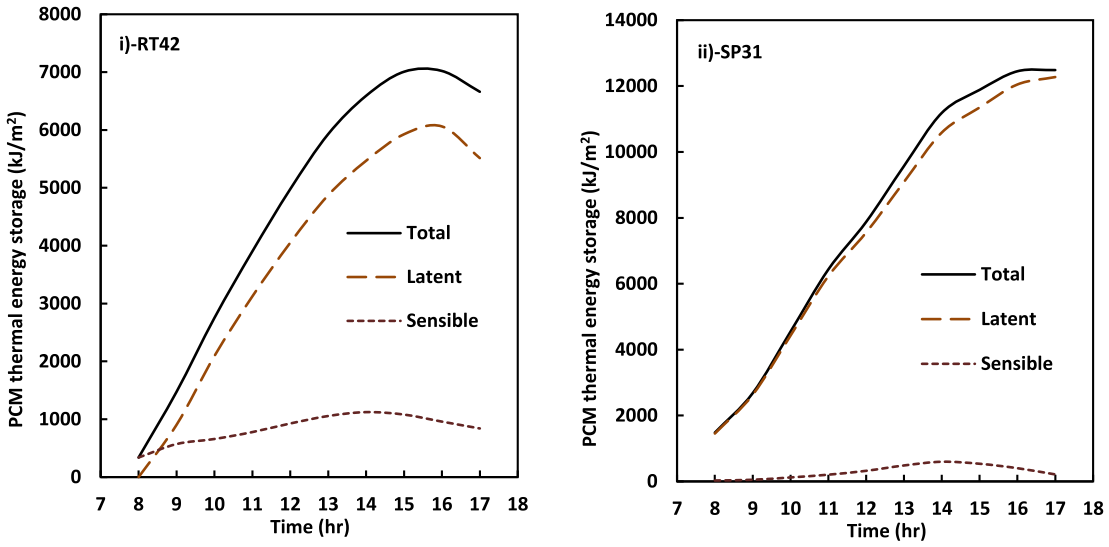
RT42 has a relatively low latent heat of fusion and a high melting point compared with other summer PCMs; accordingly, RT42 has the lowest energy stored. By the day's end, RT42 loses part of its storage heat as RT42 goes on the cooling process, so it has a negative heat storage value by the end of the thermal regulation period of about −360 kJ/m^2 (negative value means the PCM losses its heat to the PV direction). The same behavior is found in winter for RT21 and RT18HC types. The results show that the maximum hourly energy stored for summer PCMs of SP31, RT35HC, and RT42 is 1887 kJ/m^2, 1489 kJ/m^2, and 1278 kJ/m^2. These results for winter PCMs of SP15-gel, RT18HC, and RT21 are 1398 kJ/m^2, 1198 kJ/m^2, and 951 kJ/m^2, respectively. These results show that the maximum hourly energy stored differs for the different PCMs depending on each PCM's melting temperature, latent heat, and incident solar radiation. For example, the thermal energy storage for SP15-gel in winter is greater than RT42 in summer despite the higher solar energy. This reveals that selecting the appropriate PCM is important for maximizing the thermal energy storage in the PCM.

Fig. 14a and b shows the variation of hourly thermal efficiency of the PCM for summer and winter conditions, respectively. The thermal efficiency of PCM is defined in Eq. (33) as the ratio between thermal energy storage inside PCM to solar radiation. From this definition, thermal energy declines as the intensity of solar radiation increases until noontime; then, in the afternoon, the efficiency drops due to the decrease in the rate of heat absorbed by PCM because most of the heat is absorbed sensibly, and PCMs have been melted. PCM types (RT42, RT21, and RT18HC) have negative thermal efficiencies at the end period of the thermal regulation of the PV (due to the back heat flow from PCM to PV

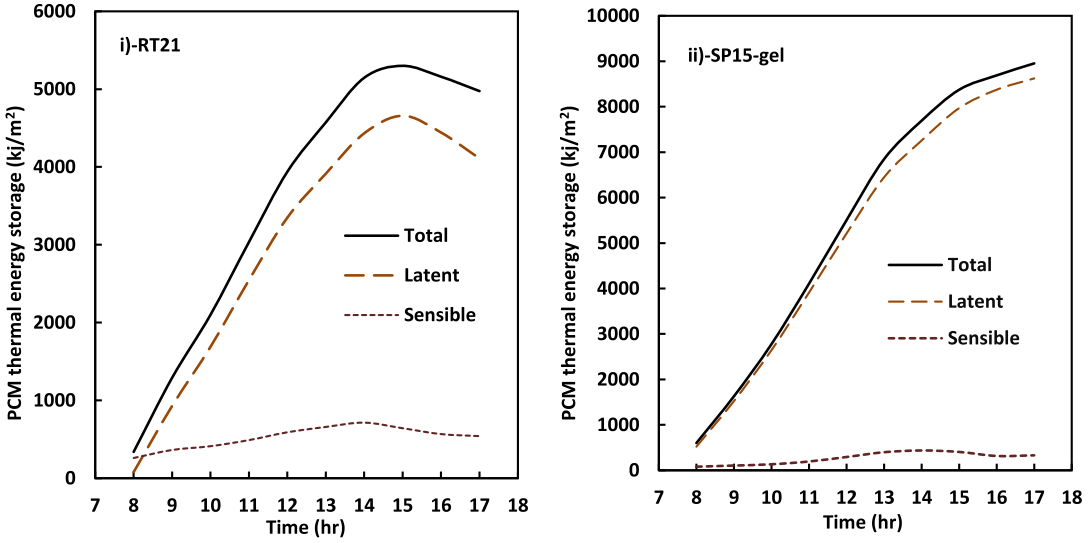


(a) Summer (b) Winter

Fig. 14. Hourly thermal efficiency of the PCMs used in (a) summer and (b) winter.



(a) Summer



(b) Winter

Fig. 15. Accumulated absorbed PCMs thermal energy in (a) summer and (b) winter.

panel, as explained early). The maximum hourly thermal efficiency for summer PCMs of SP31, RT35HC, and RT42 is 54.7%, 43.7%, and 42%, respectively. These results for winter PCMs of SP15-gel, RT18HC, and RT21 are 57%, 53.3%, and 51.8%, respectively.

PCM absorbs heat in the form of sensible and latent heat, according to Eq. (34). Fig. 15a and b shows the accumulated heat storage in PCM types for summer and winter conditions. From these figures, a significant percentage of total heat storage in PCM is latent heat, with a small portion stored sensibly. Initially, both latent and sensible heat storage increase with time because PCM is firstly heated sensibly until reaching the melting point, and then latent heat dominates the energy storage. Sensible energy increase because the melted-liquid PCM is also heated sensibly until 14:00. After that, accumulated sensible energy declines as hotter PCM layers lose their sensible heat to melt the remaining solid part, and solar intensity decreases. The accumulated thermal energy storage for PCMs having a low melting point (SP15-gel, SP31, and RT35HC) rises continuously until the end of the day because backflow heat does not occur. RT42 has sensible heat greater than latent heat initially because RT42 has a high melting point of 37 °C. As a result, a small amount of PCM is melted, and a large portion of heat is absorbed sensibly for the first hour.

Fig. 15ai shows that accumulated latent heat decreases as part of PCM is solidified, as explained before. The behavior of RT42 is similar to RT21 used for winter conditions. Fig. 15a case SP31 illustrates that sensible heat is mergeable compared with latent heat for the first hour. The reason for that is that SP31 starts to melt latently, and at the same time, the melted PCM is heated sensibly. The results show that the total daily accumulated solar for summer PCMs SP31, RT35HC, and RT42 is 12485 kJ/m^2, 8574 kJ/m^2, and 6660.7 kJ/m^2 of the PV surface area, respectively, while these results for winter PCMs SP15-gel, RT18HC, and RT21 are 8953 kJ/m^2, 6480.8 kJ/m^2, and 4975 kJ/m^2, respectively.

Fig. 16a, b, and 15c, respectively, illustrate the average daily electric efficiency of PV cell, thermal efficiency, and PV/HP-PCM system efficiency for all studied PCM types in winter and summer compared with

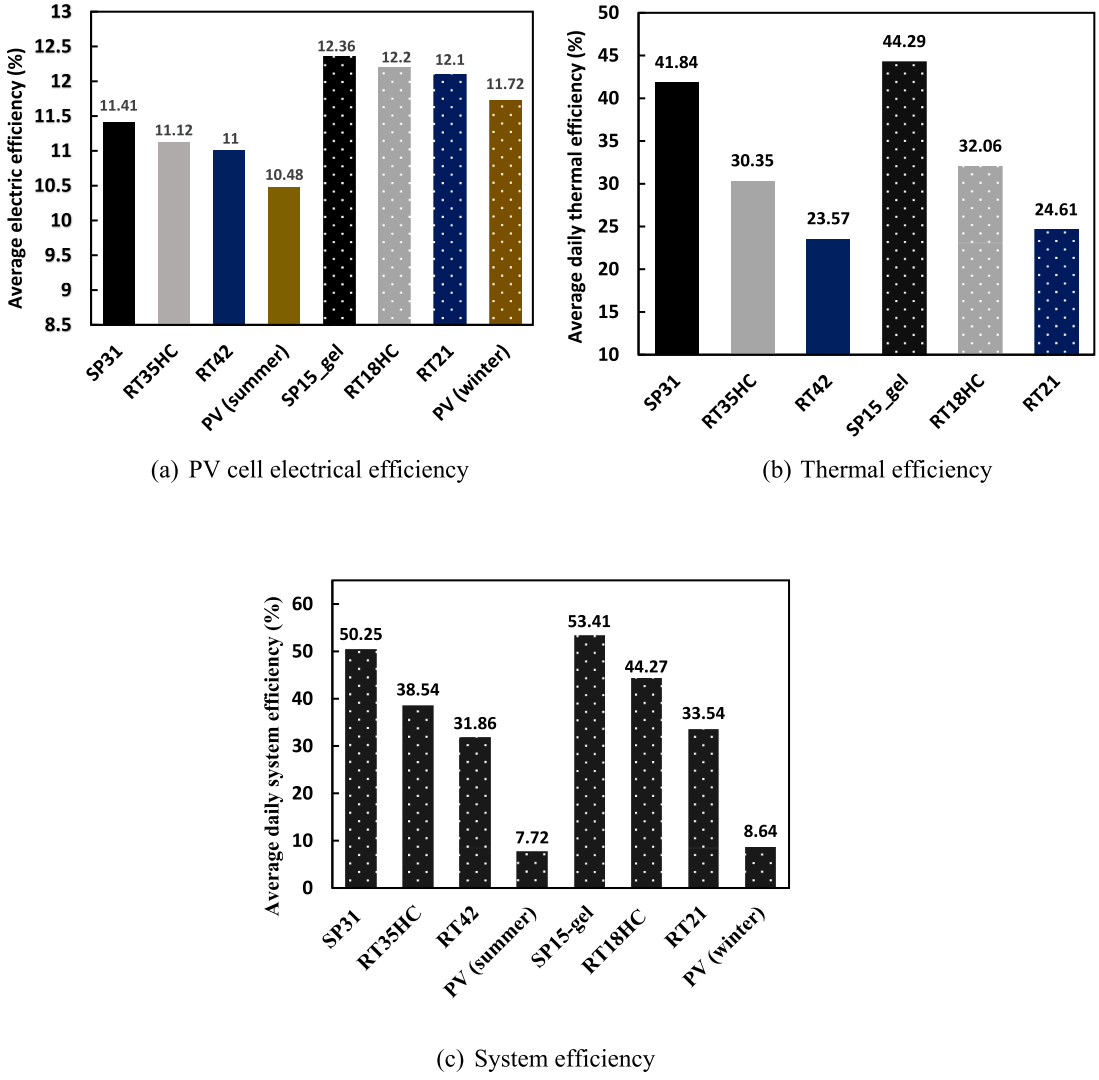


Fig. 16. Average daily (a) electric efficiency of PV cell, (b) system thermal efficiency, and (c) total system efficiency compared with conventional PV efficiency.

conventional PV panels efficiency. As explained before, SP31 and SP15-gel have the maximum reduction of cell operating temperature, so they have the highest average daily electric and thermal efficiencies. Fig. 16a shows that all studied PV/HP-PCM cases have higher electrical PV efficiency than conventional PV. They also have higher daily efficiency, as stated in Fig. 16c. The results also reveal that the average daily thermal efficiency is higher than the average electrical efficiency, which indicates that the portion of the solar energy that transfers to thermal energy storage in the PCM is higher than the portion of the solar energy that transfers to electrical power.

It is also clear that the average PV electrical efficiency in winter is higher than in summer because of the lower PV temperature. For example, the average daily electrical efficiency compared to thermal efficiency is 25.5% for SP31 in summer and 26.5% for SP15-gel in winter. The maximum increase of the PV cell efficiency compared to conventional PV in summer is about 8.8% and about 5.4% in winter. The present results are compared with the results of (Elsheniti et al., 2020) for the thermal regulation of the PV with PCM in winter conditions. It is found that the enhancement of the average electrical efficiency of the current work is about 5.4% compared with 5% obtained by Elsheniti et al. Moreover, compared with the results obtained by (S. A. Nada, D. H. El-Nagar, 2018) for summer conditions, the present enhancement is 8.8% compared with 5.7% obtained by Nada. This comparison indicates that the current system is more performant in PV thermal regulation than the previous studies.

It is noted that the electrical efficiency of the PV cell is relatively small depending on the type of the used cell in the analysis. By selecting a more efficient PV, the impact of the cooling system on the PV efficiency will increase also. The average daily system efficiency for the best winter PCM type, SP15-gel, is 53.41%, slightly higher than that for the better summer PCM type SP31 of 50.25%. This difference is because winter types have higher electrical efficiency than summer types. The results show that for the impact of PCM type, the maximum rise of the average daily PV electric, thermal, and system efficiencies for all summer PCM materials is 3.7%, 77.5%, and 57.7%, respectively, while for winter PCMs cases is 2.1%, 80%, and 59.2%, respectively. That means PCM type has a more significant impact on the electrical PV efficiency in summer than winter. In contrast, it has a higher effect on the thermal efficiency in winter than in summer.

6. Exergy analysis

6.1. PV exergy efficiency

Fig. 17 illustrates the evolution of the exergy efficiency with the time of the hybrid PV system compared with natural convection cooled PV for both summer and winter conditions. The exergy efficiency profile for all PV cases for both climate conditions has the same electrical efficiency

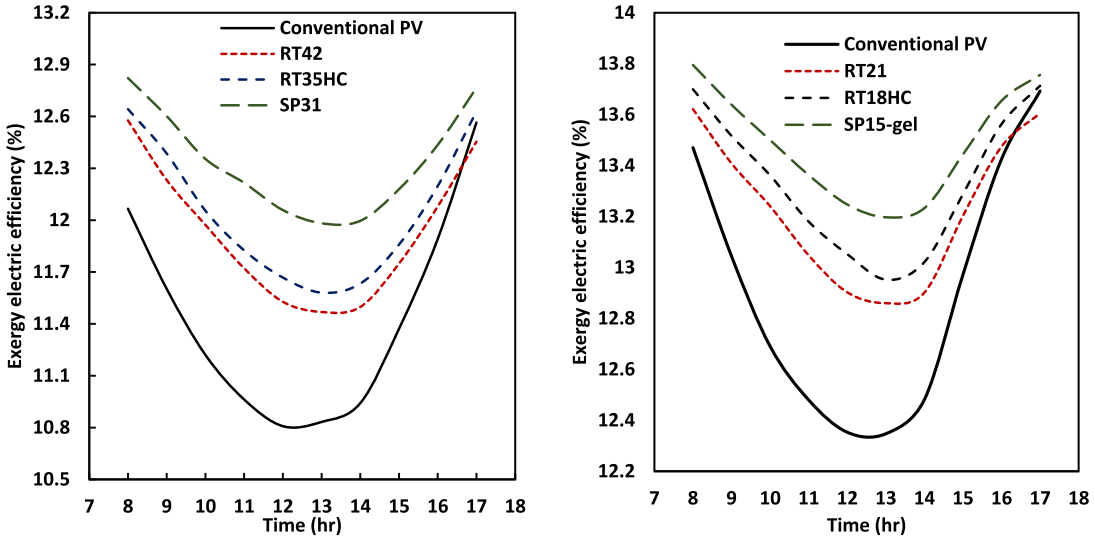


Fig. 17. Variation of exergy efficiency of the PV cell in (a) summer and (b) winter.

variation trend. This profile is because their electrical performance strongly influences the exergy efficiency of PV systems. For both summer and winter, the exergy efficiency values fall gradually from the initial until reaching their lowest point at noontime before soaring again for the rest of the day until sunset. Compared with the reference PV cell, exergy efficiency for the hybrid cooling system is high. The heat removed by the HP-PCM cooling system may be to blame for this since it lowers the panels' temperature and increases their electrical output. For summer PCM types, it is found that SP31 has higher exergy efficiency profile compared with other summer types due to the reduction in cell temperature. Both RT42 and RT21 have exergy efficiency lower than the conventional PV panel at 16:00 due to backflow heat.

6.2. PCM exergy efficiency

Fig. 18 depicts the hourly variation of PCM exergy efficiency with time during the thermal regulation of the PV system. The results of the summer exergy are shown in Fig. 18a, whereas the findings for the winter day are revealed in Fig. 18b. The exergy efficiency of RT42 is higher than RT35HC and SP31 in summer because RT42 stores heat at a high temperature relative to the environment's temperature. It is worth mentioning that the exergy efficiency's negative sign means the energy returns from the PCM to the PV direction, as explained previously. The maximum hourly PCM exergy efficiency is accounted for RT42 of 1.18%, compared with 0.62% and 0.31% for RT35HC and SP31, respectively. The exergy efficiency for winter PCM types experiences the same behavior in the summer type, where rising the melting point increases the exergy efficiency. Fig. 18b illustrates that RT21 achieves maximum exergy efficiency of 1.23% compared with 0.77% and 0.74% for RT18HC and SP15-gel, respectively.

Fig. 19a and b illustrate the average daily exergy efficiency of the PV panel and PCM for different studied PCMs, respectively. Winter's average PV exergy efficiency is higher than the summer case. Fig. 19a shows that in winter, the average PV exergy efficiencies for uncooled PV, RT21, RT18HC, and SP15-gel were 12.71,13.12%, 13.22%, and 13.4%, respectively, whereas, for summer conditions, the average exergy efficiencies were 11.25%, 11.81%, 11.94%, and 12.25% for reference PV, RT42, RT35HC, and SP31, respectively. Fig. 19b shows that in winter, the average daily exergy efficiency of the PCM is about 0.7, 0.5, and 0.38 for RT21, RT18HC, and SP 15-gel, respectively, while in summer is 0.69, 0.52, and 0.15 for RT42, RT35HC, and SP31, respectively. By comparing the exergy results, it is noted that the exergy efficiency of PV is higher

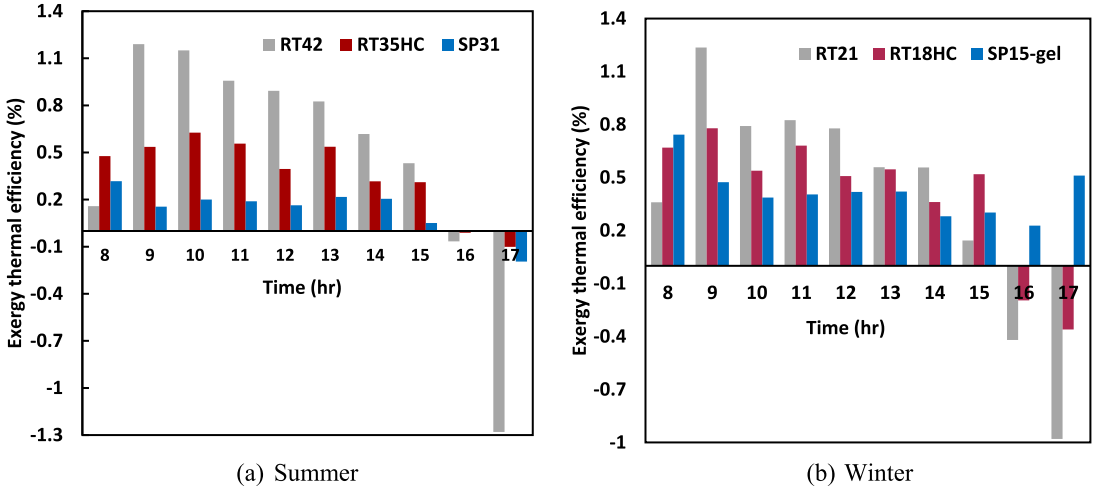
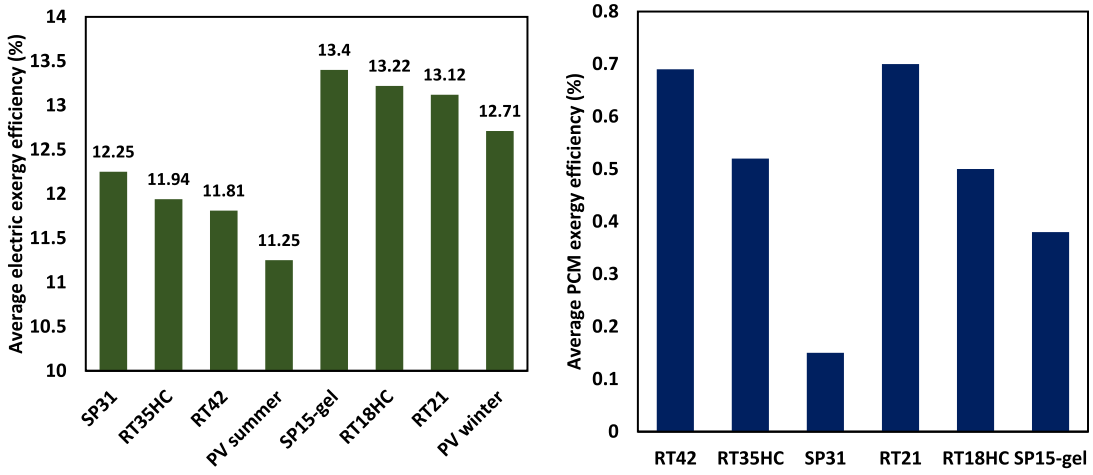


Fig. 18. Variation of exergy efficiency of the PCMs used in (a) summer and (b) winter.



(a) PV average exergy efficiency (b) PCM average exergy efficiency

Fig. 19. The average daily exergy efficiency of (a) PV modules and (b) PCM.

Table 7
Capital cost of each item of the hybrid cooling system.

Item	CC
PV Module (20 mm × 25 mm), P_{max} = 5W	10.16 $
Heat pipes (500 mm × 50 mm)	12 $
PCM	6.6 $

than that of PCM because the PV produces electrical energy while PCM produces heat energy.

7. Cost analysis

Table 7 shows the capital cost of PV panels, heat pipes, and PCM, while Table 8 presents the cost analysis results of the proposed hybrid cooling system and reference PV. The analysis disclosed that the cost of electricity production from the hybrid PV/HP-PCM system is lower than using the natural PV cooling case. For the hybrid PV system (PV/HP/PCM), the lifetime is taken as 20 years, while the interest rate is taken as 9.75%, according to the Egyptian Central Bank. Moreover, the annual electricity production, E_{annual} (kWh/year) equal the direct electric power generation from the PV panel plus energy generated from thermal energy stored in PCM with a conversion efficiency of 35%. As summarized in Table 7, the cost of 1 kW from the hybrid PV system equals 0.1257 $, compared with 0.1341 $/kWh for the conventional PV cell. This cost comparison shows that a 6.26% reduction in electricity cost could be achieved using the hybrid PV/HP-PCM system.

Fig. 20 illustrates the effect of altering the interest rate and lifespan of the photovoltaic system on the electricity production cost. Three different interest rates of 4%, 8%, and 12% are used for three different PV lifetimes of 10, 20, and 30 years. Energy production cost decreases with increasing the PV lifetime, while it rises with increasing the interest rate. Furthermore, increasing the interest rate has a significant impact on increasing the energy cost compared with reducing it by increasing PV lifetime. Fig. 20 disclosed that the change in electricity cost is approximately constant by increasing the system lifetime from 20 to 30 years. For example, the reduction in energy cost is 6.8% for PV lifetime change from 20 to 30 years, compared with a reduction of 23% from 10 to 20 years at an interest rate of 12%. Fig. 21 illustrates the variation of cumulated cash flow for 20 years lifespan of the hybrid PV system and the unmodified PV panel. Fig. 21a shows that the payback period for the conventional PV panel is about 15.01 years compared with only 13.67 years for the PV/HP-PCM system, as shown in Fig. 21b.

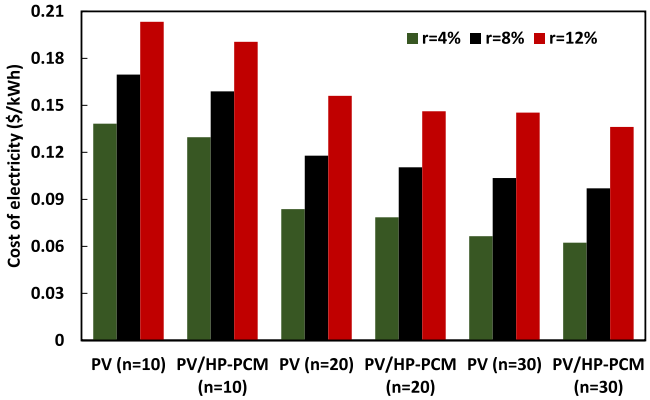


Fig. 20. Cost of energy generation at different interest rates and PV lifetimes.

8. Conclusions

This paper has introduced a new PV panel cooling system design by integrating the photovoltaic cell with heat pipe and phase change material (PV/HP-PCM). The impact of PCM type and quantity has been investigated on the electric and thermal performance of PV (cell temperature, energy and exergy efficiency, and output power) and PCM thermal performance (temperature evolution, liquid fraction, energy and exergy efficiency, and energy storage). The study is conducted for three PCMs in summer; RT42, RT35HC, and SP3, and three in winter; RT21, RT18HC, and SP15-gel. Moreover, cost analysis is investigated for

Table 8
Cost analysis of the studied PV system.

System	CC($)	CRF ($)	FAC ($)	AMC ($)	S ($)	SSF ($)	ASV ($)	TAC ($)	E $_{annual}$ (kWh/year)	Pe ($/kWh)
PV	10.16	0.115	1.173	0.1760	0.76	0.018	0.0137	1.335	9.955	0.1341
PV/HP-PCM	28.76	0.115	3.32	0.4981	2.15	0.018	0.0387	3.78	30.07	0.1257

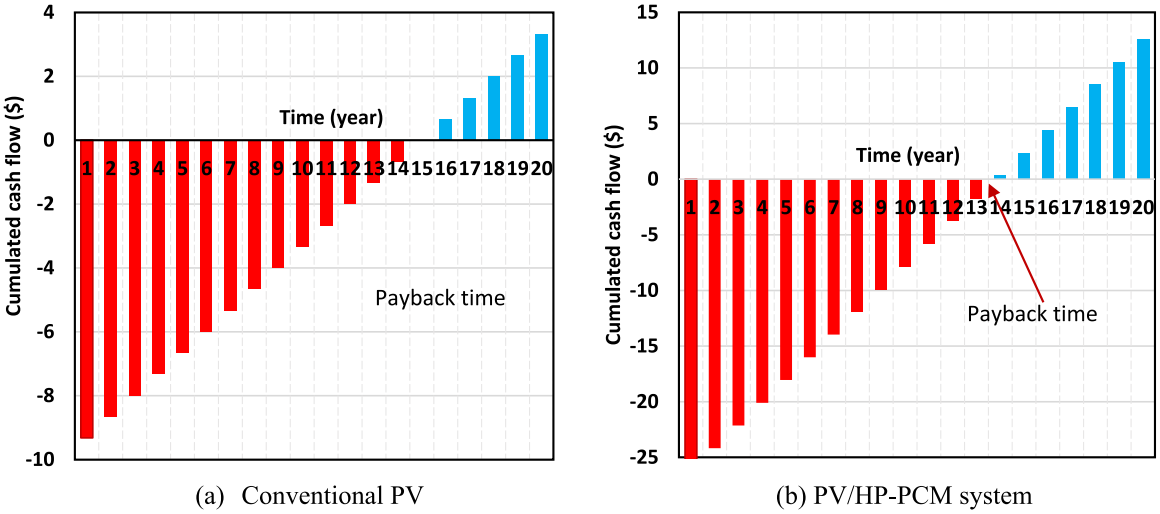


Fig. 21. Variation of cumulated cash flow for (a) Conventional PV and (b) PV/HP-PCM system during 20 years PV lifetime.

energy production. The main conclusion of the work results can be summarized as follows:

- In summer, the maximum reduction in cell temperature accounted for SP31 is 20.6 °C at PCM best thickness of 3 cm, compared with 11.9 °C and 14.2 °C of RT42 and RT35HC, respectively. Winter data show that using SP15-gel achieves a maximum reduction of 14.1 °C at 2.5 cm thickness, compared with 9.5 °C, 10.1 °C for RT21 and RT18HC, respectively.
- The hybrid cooling system enhances the electric performance of PV/HP-PCM compared with the conventional PV panel, where the enhancement using RT42, RT35HC, and SP31 is 6.6%, 7.9%, and 11.5%, respectively. At the same time, RT21, RT18HC, and SP15-gel achieve 4.4%, 5.6%, and 7.2%, respectively.
- The exergy efficiency of the PV is greater than that of PCM. In addition, with increasing phase change temperature, the PCM exergy efficiency rises whereas PV exergy efficiency plummets. In winter, SP15-gel achieved maximum daily PV exergy efficiency of 13.4%, compared with 12.71% of uncooled PV, while SP31 achieved 12.25%, compared with 11.25% of conventional PV in summer.
- The cost of electricity production of the hybrid PV/HP-PCM system is 0.1257$/kWh, compared with 0.1257$/kWh for the conventional PV cell. Also, the PB for the hybrid system and conventional PV panel is 13.67 and 15.01 years, respectively, with an interest rate of 9.75%.
- As a result of the efficient performance of the PV/HP-PCM cooling system, the performance of the proposed cooling system for regular and concentrated photovoltaic cell could be further deployed in experimental and theoretical future studies. Also, the performance of the current system using only one PCM or two different PCM (summer and winter) through the whole year could be investigated

CRediT authorship contribution statement

Ramadan Gad: Conceptualization, Methodology, Investigation, Validation, Writing – original draft, Writing – review & editing. **Hatem Mahmoud:** Supervision. **Shinichi Ookawara:** Supervision. **Hamdy Hassan:** Conceptualization, Methodology, Investigation, Writing – review & editing, Supervision.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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